COURSE SLIDES · WSQ

Cisco Certified Network Professional (CCNP) for ENCOR Training

WSQ Course Code: TGS-2024044052

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v6.0 · 7 September 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE
Enterprise networking, Cisco routing and switching, network security and network automation.

DELIVERS
WSQ courses on CCNA, CCNP Enterprise, network security and infrastructure automation.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- Your networking experience — CCNA, hands-on with Cisco IOS, or new to enterprise networking.
- The network you look after today: campus, branch, data centre or wireless.
- What you want to be able to configure or troubleshoot by Friday.

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HOUSEKEEPING

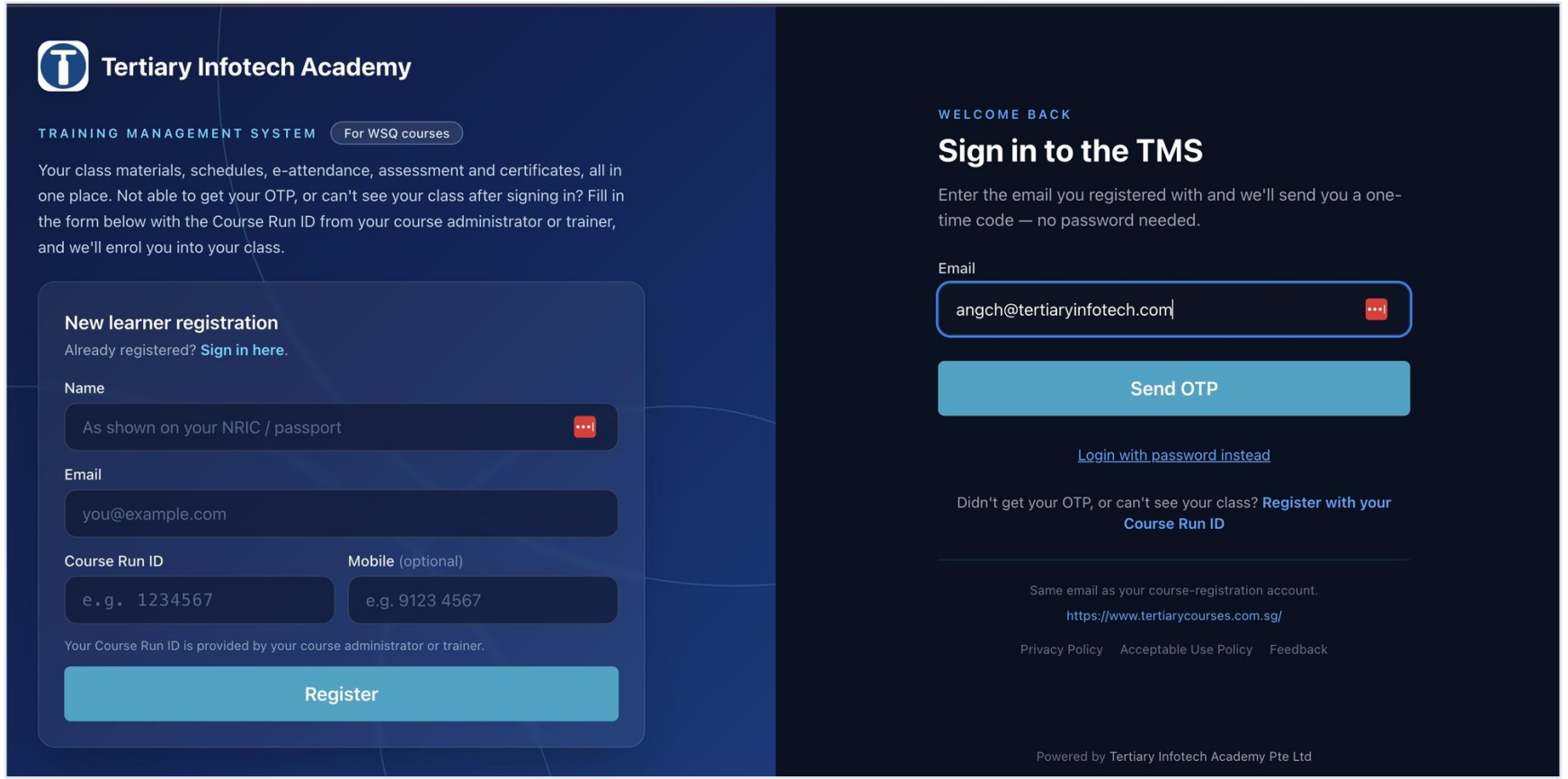
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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LMS / TMS · lms-tms.tertiaryinfotech.com

Download Your Course Material

1

1 · Go to the LMS portal

Open https://lms-tms.tertiaryinfotech.com in your browser.

2

2 · Log in

Sign in with the account e-mail you used to register for this course.

3

3 · Open this course

Select 'Cisco Certified Network Professional (CCNP) for ENCOR Training'.

4

4 · Download

Download the Trainer Slides, Learner Guide and Lesson Plan (PDF).

5

5 · Get the labs

Open the labs folder link on the course page for every Packet Tracer topology.

6

6 · Keep them open

You may use these materials during the open-book assessment.

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LMS / TMS · STEP BY STEP

How to Download Your Course Material

1

Open the LMS

lms-tms.tertiaryinfotech.com

2

Log in

Your registration e-mail

3

Find this course

CCNP ENCOR · TGS-2024044052

4

Open Courseware

Slides · LG · LP · Labs

5

Download the PDFs

Keep them for the open book

BEFORE WE START

Download all four documents now — the assessment is open book, but only these approved materials are permitted.

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Skills Framework Alignment

1

TSC Title

Network Configuration

2

TSC Code

ICT-DIT-3009-1.1

3

K1 / A1

Key network components and their interdependencies; translating organisational to technical requirements.

4

K2 / A2

Techniques for implementing and configuring servers, devices and network components.

5

K3 / A3

Features of network management tools; implementing devices per the network blueprint.

6

K4 / A6

Configuration of network servers and devices; procedures for reporting and stability assessment.

7

K5 / A4

Network access management; managing user network access to organisational guidelines.

8

K6 / A5

Network tracking and assessment tools; identifying suitable network management tools.

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Translate requirements

Translate organizational networking needs into technical configurations by selecting appropriate network components, considering their functions and interdependencies.

2

LO2 · Select components

Identify and choose key network components for configuration, utilizing techniques in server and device implementation aligned with technical requirements.

3

LO3 · Implement devices

Implement a variety of servers and devices effectively, following the organizational network blueprint to enhance network performance and reliability.

4

LO4 · Manage access

Manage network access, ensuring compliance with organizational guidelines and specifications to maintain secure and efficient user connectivity.

5

LO5 · Deploy tools

Select and deploy network management tools effectively, optimizing network performance and user experience in line with organizational needs.

6

LO6 · Report and assess

Conduct regular network reporting and administration, employing assessment tools to ensure network stability and alignment with organizational objectives.

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SIX TOPICS · 35 HANDS-ON LABS

Course Outline — The Six ENCOR Exam Domains

1

Topic 01 — Enterprise Architecture · 15%

Design principles · wireless design · SD-WAN · SD-Access · QoS · switching mechanisms

2

Topic 02 — Virtualization Technologies · 10%

Hypervisors · virtual machines · virtual switching · VRF · GRE and IPsec · LISP and VXLAN

3

Topic 03 — Network Infrastructure · 30%

Layer 2 · Layer 3 routing · wireless · IP services — the largest exam domain

4

Topic 04 — Network Assurance · 10%

Diagnostics · NetFlow · SPAN/RSPAN/ERSPAN · IP SLA · DNA Center · NETCONF and RESTCONF

5

Topic 05 — Network Security · 20%

Device access control · AAA · ACLs and CoPP · REST API security · wireless security · security design

6

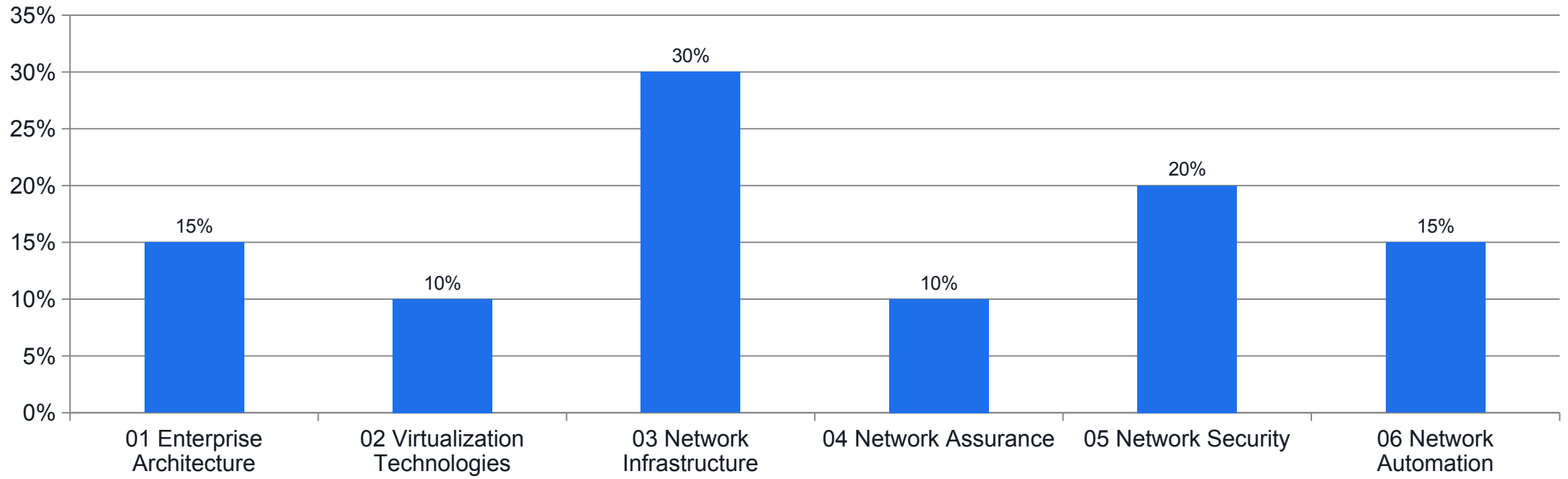
Topic 06 — Network Automation · 15%

Python · JSON · YANG · DNA Center and vManage APIs · REST responses · EEM · orchestration tools

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ENCOR 350-401 · BLUEPRINT WEIGHTING

Where Course Time Goes — Exam Domain Weighting



WHAT THE DATA SHOWS

Infrastructure carries 30% of the exam — nearly a third — and Security another 20%. Half the exam sits in these two domains, so half your revision time belongs there. Days 3 and 4 are built around that.

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SCHEDULE · 40 INSTRUCTIONAL HOURS

Lesson Plan — 5 Days, 8 hours/day

Days 1–3

- Day 1 — Enterprise Architecture — design, switching paths, campus LAN and high availability
  - Digital Attendance (AM) · Introductions · Learning Outcomes
  - Topic 1 Enterprise Architecture (Labs 1–5)
- Day 2 — Virtualization Technologies — hypervisors, VRF, GRE/IPsec, LISP and VXLAN
  - Topic 2 Virtualization Technologies (Labs 6–9)
- Day 3 — Network Infrastructure — Layer 2, EIGRP/OSPF/BGP and IP services
  - Topic 3 Network Infrastructure — Layer 2 and Layer 3 (Labs 10–18)

Days 4–5

- Day 4 — Network Infrastructure (wireless), Network Assurance and Network Security
  - Topic 3 Network Infrastructure — wireless and IP services (Labs 19–21)
  - Topic 4 Network Assurance (Labs 22–25)
  - Topic 5 Network Security (Labs 26–30)
- Day 5 — Network Automation, revision and final assessment
  - Topic 6 Network Automation (Labs 31–35)
  - Revision · TRAQOM Survey · Digital Attendance (Assessment)
  - Final Assessment (WA 1 hr + PP 1 hr)

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SET UP BEFORE WE START

Your Lab Environment

1

Cisco Packet Tracer 8.2+

The primary lab environment. Every .pkt/.pka topology in labs/ opens in Packet Tracer — free from the Cisco Networking Academy.

2

Cisco IOS / IOS XE CLI

All configuration is done at the IOS CLI, so the commands transfer directly to real 9200/9300 switches and ISR/CSR routers.

3

Wireless LAN Controller (WLC)

The Packet Tracer WLC GUI is used for the WLAN, WPA2-Enterprise and wireless troubleshooting labs.

4

Python 3.10+

Used for the automation labs — interpreting scripts, parsing JSON and calling REST APIs.

5

Postman or curl

Used to exercise the Cisco DNA Center and RESTCONF APIs and to read response codes.

6

A text editor

VS Code or any editor for writing Python scripts, JSON payloads and saved configurations.

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BEFORE YOU START

Briefing for Assessment

1

Do · Clear your desk

Place phones and other materials under the table or on the floor.

2

Don't · No recording

No photos or recording of assessment scripts.

3

Don't · No discussion

Work individually — no discussion during the assessment.

4

Do · Black or blue pen

Use a black or blue pen for hard-copy assessments.

5

Don't · No correction fluid

No liquid paper or correction tape may be used.

6

Do · Stop on time

Scripts are collected when time is up.

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FINAL ASSESSMENT

Final Assessment

1

Written Assessment (WA)

Short-Answer Questions (SAQ) · 1 hour · open book. Tests the underpinning knowledge taught in the slides.

2

Practical Performance (PP)

Hands-on Cisco configuration and verification tasks · 1 hour · open book. Tests the ability you built in the labs.

3

Open book

You may use the course slides, the Learner Guide and approved materials only.

4

Eligibility

A minimum of 75% attendance is required to be eligible for assessment and funding.

5

Result

You are assessed as Competent (C) or Not Yet Competent (NYC) on each instrument.

6

Appeals

An appeal process is available if you wish to contest an assessment outcome.

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ON ASSESSMENT DAY

Assessment Flow

1

TRAQOM survey

Scan the QR code on the LMS

2

Digital attendance

Scan the SSG assessment QR

3

Sit WA then PP

Open book · 1 hour each

4

Submit on the LMS

Upload your completed answers

5

Sign the record

Sign the Assessment Summary Record

REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

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WSQ FUNDING

Criteria for Funding

1

Attendance

A minimum attendance rate of 75%, based on the SSG Digital Attendance record.

2

Assessment

Complete both assessment components and be assessed as 'Competent'.

3

Digital attendance

Scan the SSG QR code for AM, PM and Assessment on every training day.

4

TRAQOM survey

Complete the mandatory TRAQOM course feedback survey on the LMS.

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PRACTICE EXAM · exams.tertiaryinfotech.com

Practice Exam — Cisco CCNP ENCOR 350-401

1

200 original questions

Two full practice exams of 100 questions each, written for this blueprint — not redistributed exam content.

2

120 minutes each

The same duration as the real 350-401 exam, so you practise the pacing as well as the content.

3

All six domains

Architecture 15% · Virtualization 10% · Infrastructure 30% · Assurance 10% · Security 20% · Automation 15%.

4

Practice mode

Answers and full explanations revealed immediately after each question — use this while you learn.

5

Exam mode

Timed, answers hidden until you submit, progress saved automatically — use this in the final week.

6

Try it free

exams.tertiaryinfotech.com/practice-exams/cisco/cisco-ccnp-encor — a free sample exam needs no payment details.

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exams.tertiaryinfotech.com/practice-exams/cisco/cisco-ccnp-encor

Practise until the pacing is automatic.

Take the practice exam in Practice mode while you learn, then in Exam mode against the clock. 100 questions in 120 minutes leaves about 72 seconds per question — the pacing is what catches people out, not the content.

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CORE CONCEPTS

The Enterprise Network Core, End to End

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WHAT YOU ARE PREPARING FOR

ENCOR is one exam across six domains.

350-401 tests whether you can implement core enterprise network technologies — dual-stack architecture, virtualization, infrastructure, assurance, security and automation. This course walks all six, in blueprint order, with a lab for every objective.

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EXAM STRUCTURE

The ENCOR 350-401 Exam at a Glance

1

120 minutes

The exam runs two hours and is the core exam for both CCNP Enterprise and CCIE Enterprise.

2

Six domains

Architecture, Virtualization, Infrastructure, Network Assurance, Security and Automation.

3

Dual stack throughout

IPv4 and IPv6 are both examinable — OSPFv3 and IPv6 addressing appear alongside their v4 equivalents.

4

Configure, verify, describe

Read the verb in each objective: 'configure and verify' means hands-on, 'describe' means conceptual.

5

Core plus concentration

Passing ENCOR alone earns the Cisco Certified Specialist – Enterprise Core certification.

6

Then a concentration

CCNP Enterprise needs ENCOR plus one concentration exam such as ENARSI.

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Read the Verb — What Each Objective Actually Asks

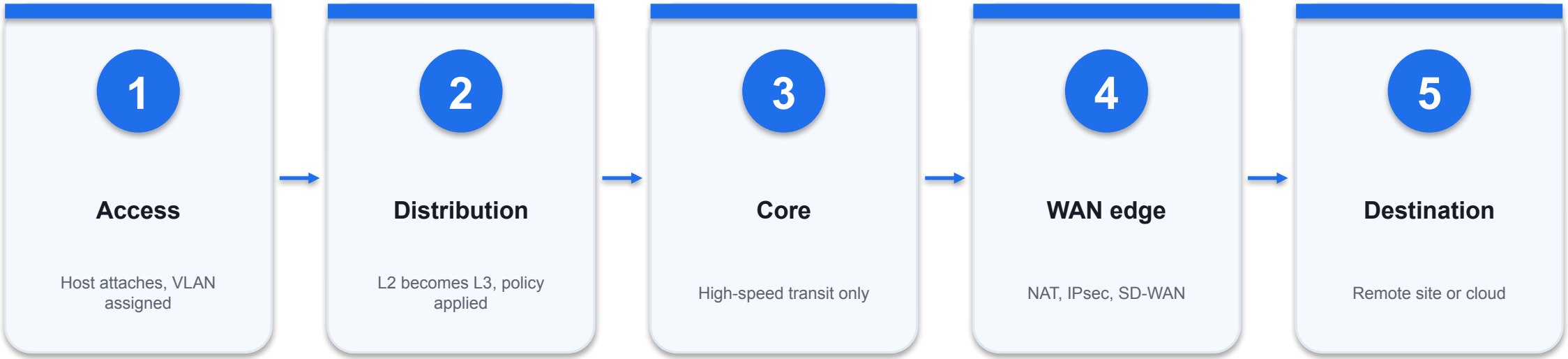
Verb in the blueprint	What the exam expects	Example objective	How this course covers it
Configure and verify	Hands-on configuration and the show command that proves it	Configure and verify eBGP	A full lab with the commands and the verification
Troubleshoot	Find and fix a broken configuration	Troubleshoot static and dynamic EtherChannels	A lab that starts broken, not blank
Describe	Explain the concept and its components	Describe REST API security	A concept walkthrough and a review worksheet
Interpret	Read a given configuration or script and predict the result	Interpret basic Python components	Predict-then-run exercises
Compare	State the differences and when to choose each	Compare agent vs agentless tools	A comparison matrix plus a justified recommendation

WHEN IT MATTERS The verb tells you how deep to go. 'Describe' will never ask you for a command; 'configure and verify' always will.

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THE JOURNEY THIS COURSE FOLLOWS

How a Packet Crosses the Enterprise Core



THE POINT

Every topic in this course is one stage of this journey. Keep this picture in mind — the exam questions are almost always about one hop on it.

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A MODEL YOU WILL USE ALL WEEK

The Three Planes — The Idea Behind Every Topic

1

Data plane

Forwards packets. CEF, the FIB and the adjacency table live here, and it runs in hardware at line rate.

2

Control plane

Decides where packets should go. Routing protocols, STP and LISP live here, and it runs in software on the CPU.

3

Management plane

How you configure and monitor the device. SSH, SNMP, NETCONF and RESTCONF live here.

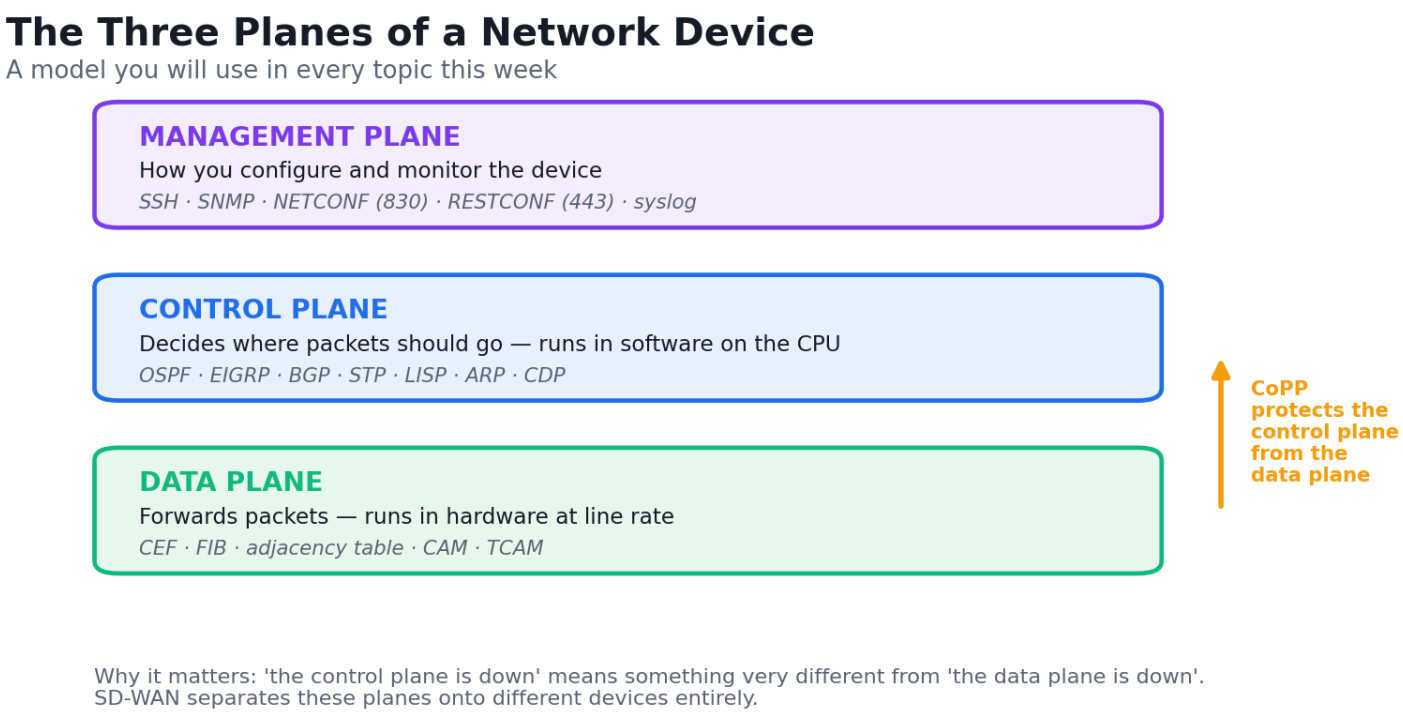
4

Why it matters

CoPP protects the control plane from the data plane, SD-WAN separates them onto different devices, and 'the control plane is down' means something very different from 'the data plane is down'.

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The Three Planes of a Network Device



Every topic this week sits in one of these three planes. Know which plane a symptom belongs to and you have already halved the troubleshooting.

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Switching Mechanisms — Process, Fast and CEF

Mechanism	How it decides	Speed	Where the decision is cached
Process switching	Every packet examined by the CPU	Slowest	Nowhere — recomputed each packet
Fast switching	First packet by CPU, rest from cache	Faster	Route cache, built on demand
CEF	FIB built ahead of any traffic	Fastest	FIB + adjacency table, in hardware

WHEN IT MATTERS CEF is the default and the answer to almost every 'which is fastest' question — because it is topology-based, not traffic-driven.

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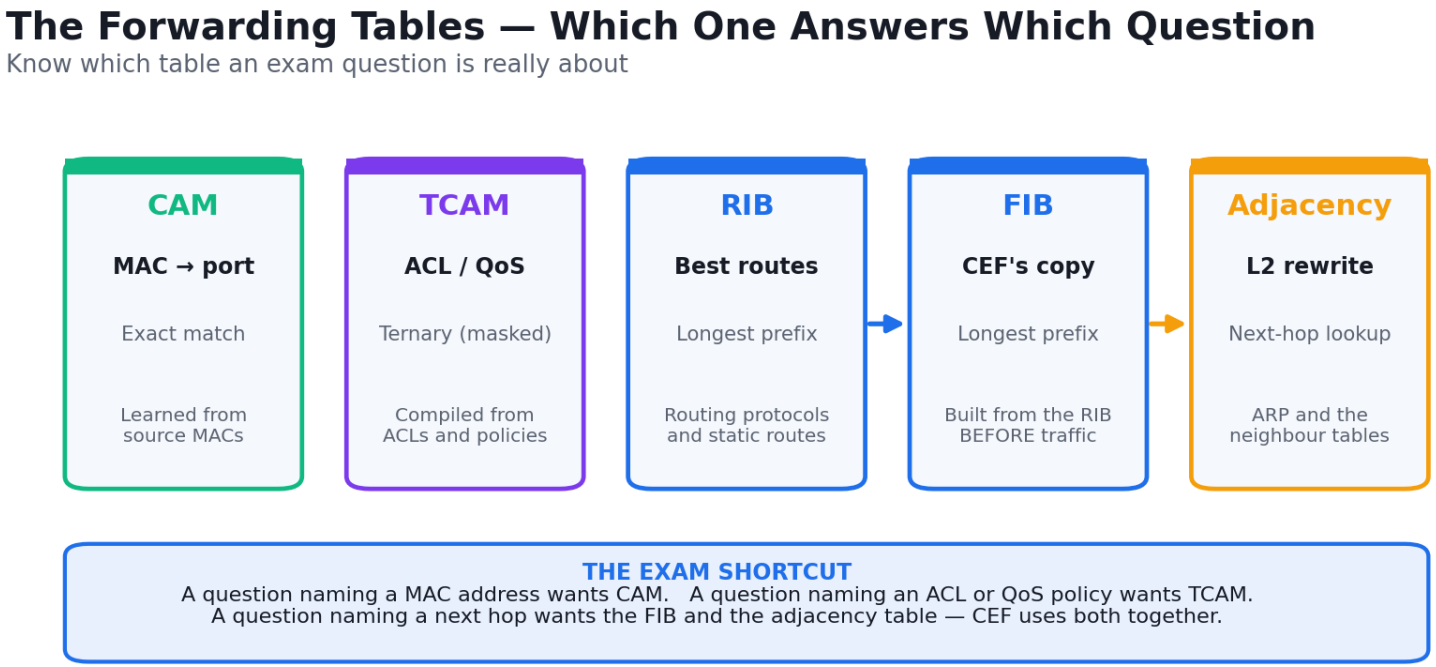
The Forwarding Tables — CAM, TCAM, RIB, FIB and Adjacency

Table	What it holds	Match type	Populated by
CAM	MAC address to port, per VLAN	Exact match	Learning source MACs from frames
TCAM	ACL, QoS and policy entries	Ternary (masked)	Compiled ACLs and QoS policies
RIB	Best routes from all sources	Longest prefix	Routing protocols and static routes
FIB	CEF's forwarding copy of the RIB	Longest prefix	Built from the RIB by CEF
Adjacency	Layer 2 rewrite for each next hop	Next-hop lookup	ARP and the neighbour tables

WHEN IT MATTERS An exam question naming a MAC address wants CAM; one naming an ACL wants TCAM; one naming a next hop wants the FIB and adjacency pair.

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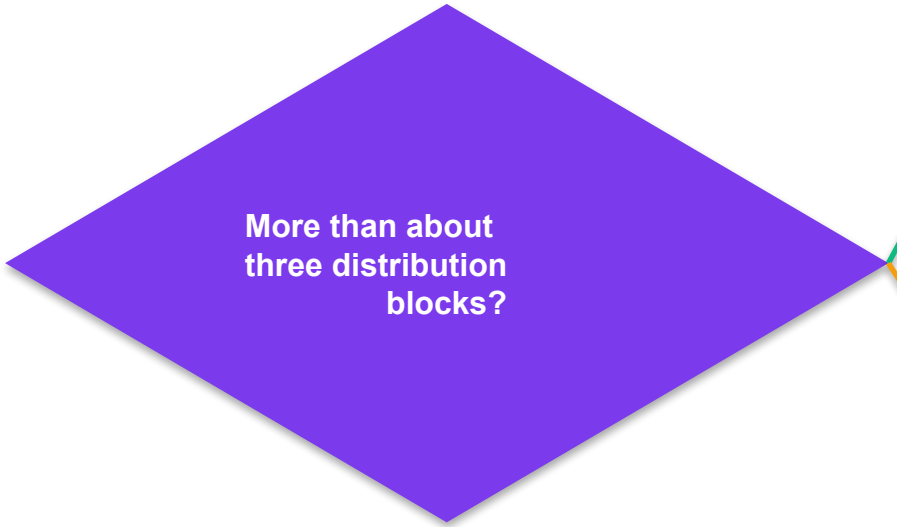
The Forwarding Tables at a Glance



CEF uses the FIB and the adjacency table together — the FIB decides where, the adjacency table builds the new frame header.

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Do You Need a Dedicated Core Layer?



YES — build a 3-tier design

Add a dedicated core. Each distribution block gets two uplinks to the core instead of meshing with every other block, so cabling and complexity grow linearly rather than quadratically.

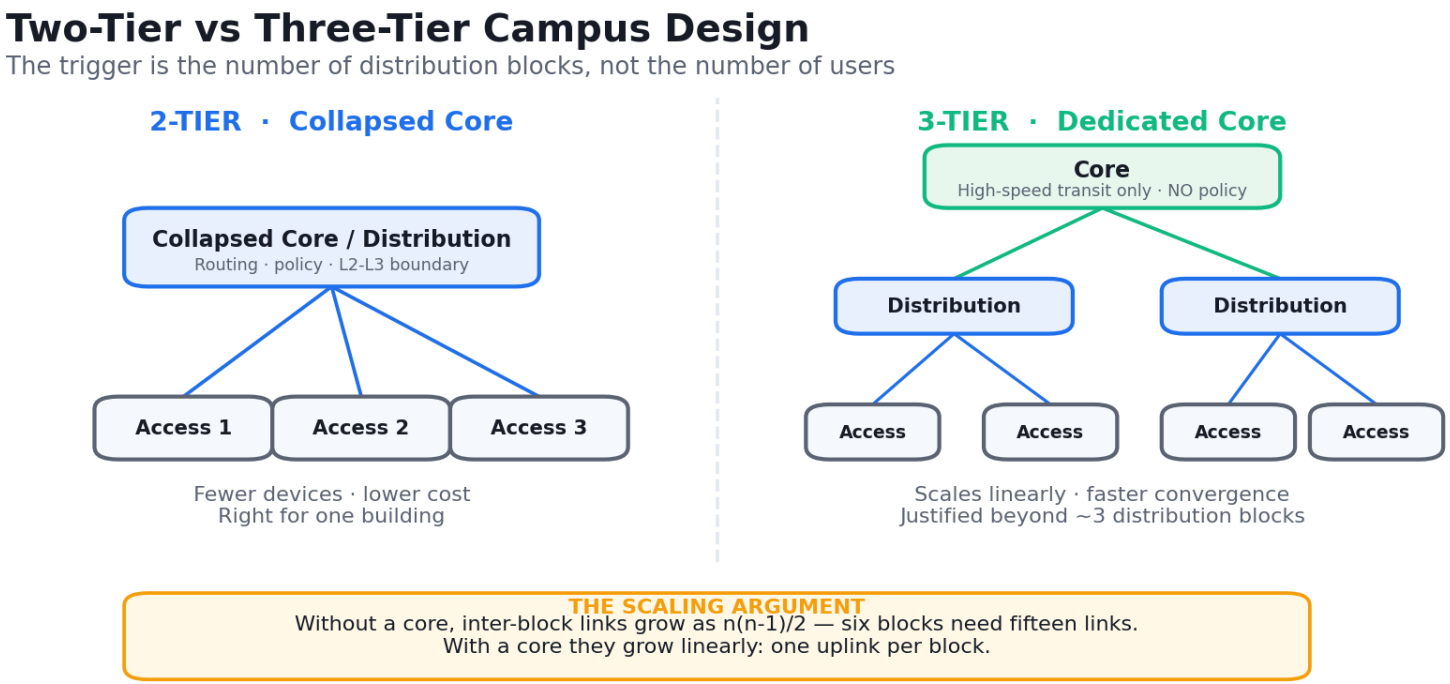
NO — collapse into 2-tier

Merge core and distribution. Fewer devices, lower cost and less to manage — the right answer for a single building or a small campus.

The trigger is the number of distribution blocks, not the number of users. Four blocks fully meshed need six links; with a core they need four uplinks.

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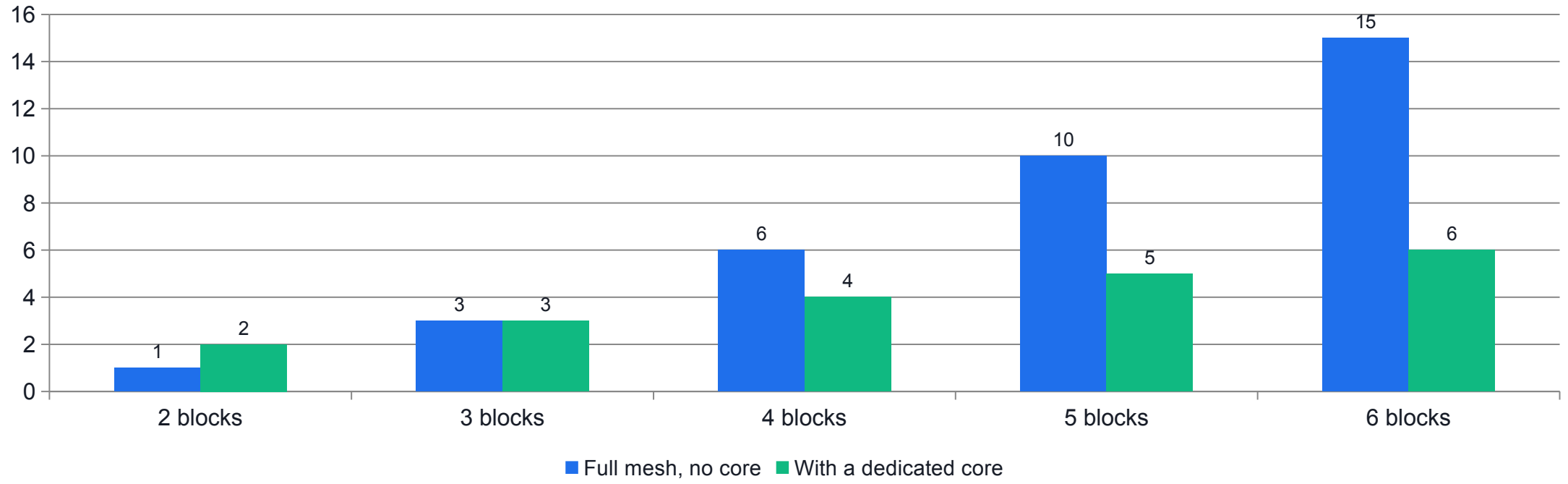
Two-Tier vs Three-Tier Campus Design



The design decision is driven by the number of distribution blocks, not the headcount. Beyond roughly three blocks, add the core.

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Why a Core Layer Scales — Links Needed as Blocks Grow



WHAT THE DATA SHOWS

Without a core, inter-block links grow as n(n-1)/2 — six blocks need fifteen links. With a core they grow linearly: one uplink per block. This is the whole quantitative case for the third tier.

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High Availability — The Three Layers of Redundancy

1

Hardware redundancy

Dual supervisors, dual power supplies and redundant line cards so a component failure is survivable.

2

Link redundancy

EtherChannel bundles and redundant uplinks so a cable or port failure does not isolate a block.

3

FHRP

HSRP, VRRP or GLBP give hosts one virtual default gateway that survives the failure of the router behind it.

4

SSO / NSF

Stateful switchover keeps the standby supervisor in sync; non-stop forwarding keeps forwarding while routing re-converges.

5

Graceful restart

Neighbours hold the adjacency during a supervisor switchover instead of tearing routing down.

6

Why all four

Each covers a different failure. Redundant hardware without FHRP still leaves hosts pointing at a dead gateway.

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First Hop Redundancy Protocols Compared

	HSRP	VRRP	GLBP
Standard	Cisco proprietary	IEEE / RFC 5798	Cisco proprietary
Roles	Active / Standby	Master / Backup	AVG / AVF
Default priority	100	100	100
Preemption	Disabled by default	Enabled by default	Enabled by default
Load balancing	No — one active router	No — one master	Yes — across all forwarders

WHEN IT MATTERS The preemption default is the classic exam trap: configure HSRP priority without 'standby preempt' and the higher-priority router will never take over.

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QoS — Mark Once, at the Trust Boundary

1

Classification

Identify the traffic — by ACL, by protocol, or by the marking a trusted device already applied.

2

Marking

Write the class into the packet: DSCP in the IP header, CoS in the 802.1Q tag. Voice is EF (46).

3

Queuing and scheduling

Decide who transmits next. LLQ gives voice a strict-priority queue; CBWFQ shares the rest by weight.

4

Congestion avoidance

WRED drops selectively before the queue is full, so TCP backs off gradually instead of all at once.

5

Policing

Enforces a rate by dropping or remarking excess — bursty, no delay added.

6

Shaping

Enforces a rate by buffering excess — smooth, but adds delay. Shape towards the provider, police from them.

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ENCOR 1.5 · WORKED EXAMPLE

Reading a QoS Policy Map

Every QoS configuration you will meet has this same three-part shape — learn to read it once.

```
class-map match-all VOICE
  match ip dscp ef
!
class-map match-all VIDEO
  match ip dscp af41
!
policy-map WAN-EDGE
  class VOICE
    priority percent 10
  class VIDEO
    bandwidth percent 30
  class class-default
    fair-queue
!
interface Serial0/0/0
  service-policy output WAN-EDGE
```

class-map

Classifies — it says WHICH packets belong to this class, and nothing else.

policy-map

Acts — it says WHAT happens to each class: priority, bandwidth or queuing.

priority percent 10

The strict-priority (low-latency) queue. Voice goes here; it is policed to 10% so it cannot starve everything else.

service-policy output

Applies the policy to an interface in one direction. Without this line the policy does nothing at all.

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Wireless Design Principles

1

Centralized

All client traffic tunnels to a central WLC over CAPWAP. Simple policy, but the WLC is in every data path.

2

Distributed / FlexConnect

The AP switches client traffic locally at the branch and only sends control traffic to the WLC — right when the WAN is slow or expensive.

3

Controller-less

Mobility Express embeds the controller in an AP. Suits a small site with no room for a separate WLC.

4

Cloud-managed

Meraki-style: management in the cloud, data stays local. Lowest operational overhead.

5

Location services

RSSI trilateration, Angle of Arrival with Hyperlocation, and BLE beacons — accuracy depends on AP density and placement.

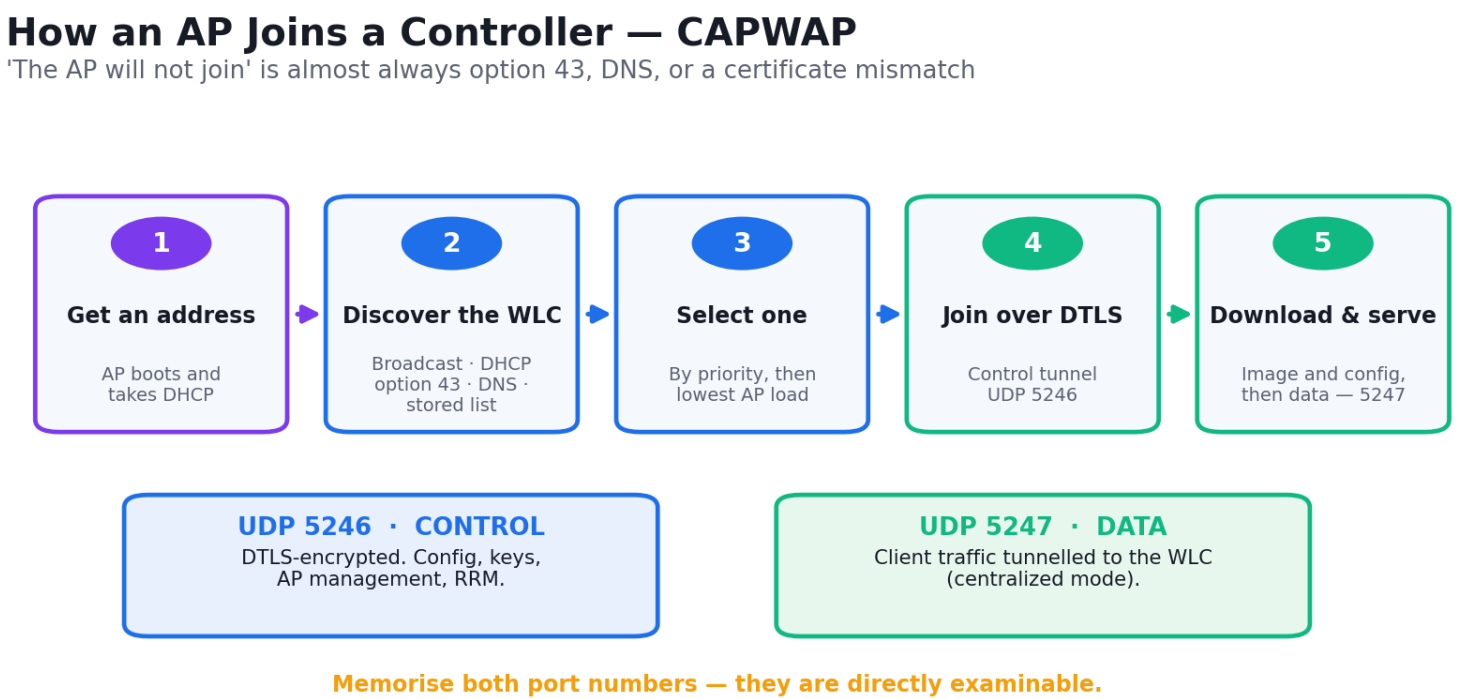
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Client density

High density is a capacity problem: smaller cells, lower transmit power, narrower channels and band steering to 5 GHz.

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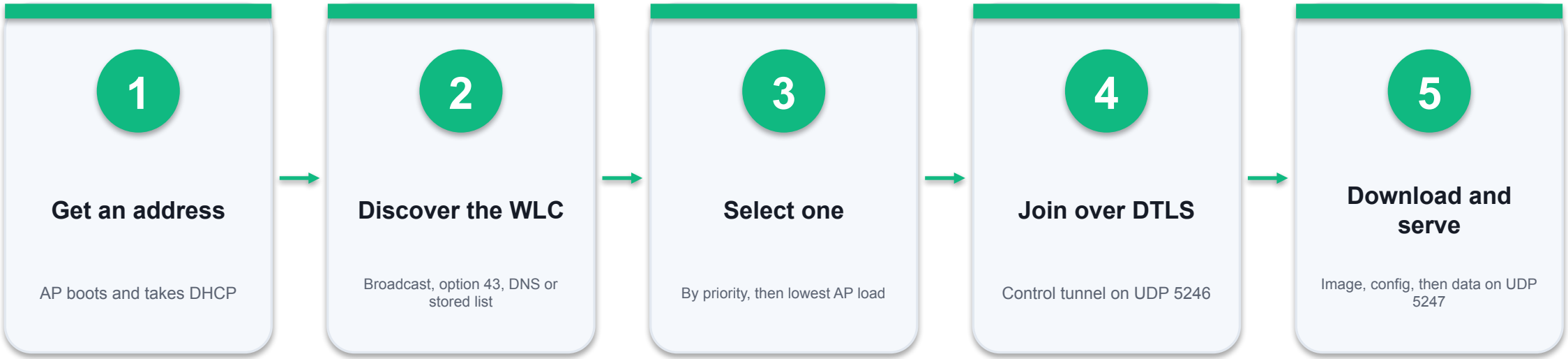
The AP Join Process — CAPWAP End to End



UDP 5246 carries the DTLS-encrypted control tunnel; UDP 5247 carries client data. Both port numbers are directly examinable.

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How an AP Joins a Controller — CAPWAP

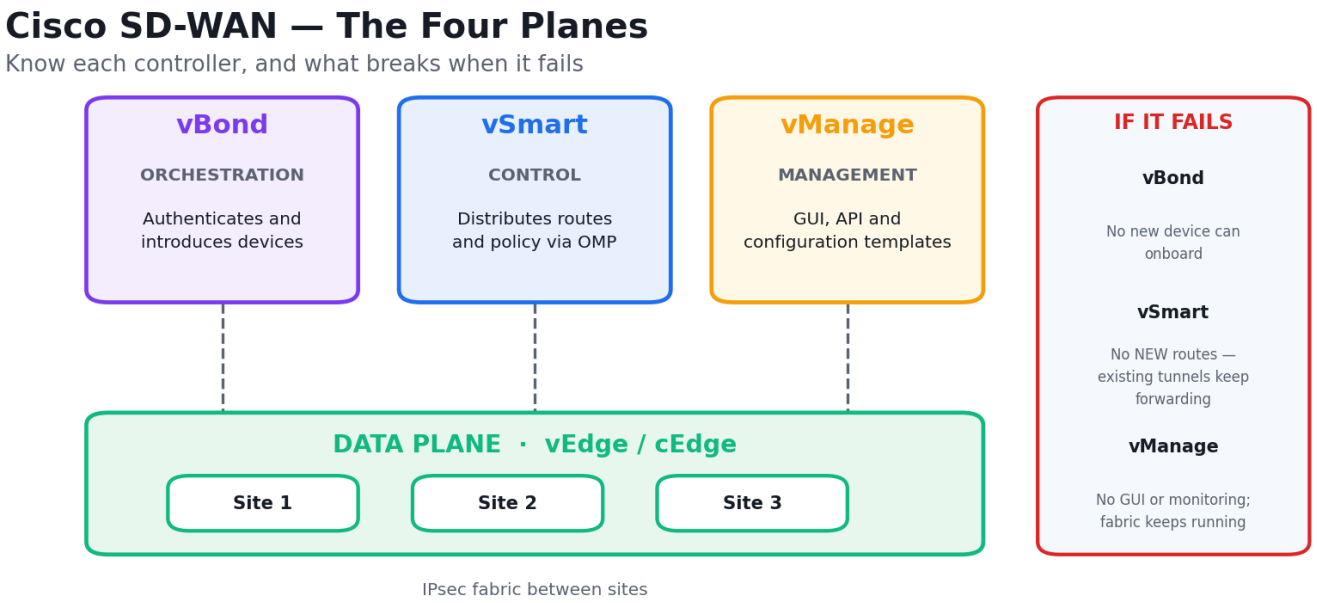


EXAM POINT

Learn the discovery order and both port numbers — 5246 control, 5247 data. 'The AP will not join' questions are almost always option 43, DNS, or a DTLS certificate mismatch.

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Cisco SD-WAN — The Four Planes and What Breaks



Exam point: only vBond must be publicly reachable. A vSmart outage stops new route learning but does NOT stop existing traffic.

> Only vBond must be publicly reachable. A vSmart outage stops NEW route learning — it does not stop traffic already flowing.

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Cisco SD-WAN — The Four Planes

1

vBond · Orchestration

Authenticates every device and introduces them to each other. The only element that must have a public address.

2

vSmart · Control

Distributes routes and policy over OMP. It never carries user data.

3

vManage · Management

The GUI, the API and the configuration templates. Where you actually operate the fabric.

4

vEdge / cEdge · Data

The routers at each site that build the IPsec fabric and forward the traffic.

5

If vSmart fails

Existing data plane tunnels keep forwarding — but no new routes or policy are learned.

6

If vManage fails

The fabric keeps running; you lose management and monitoring, not forwarding.

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Cisco SD-Access — Underlay, Overlay and Policy

1

Underlay

Any routed IP network that provides reachability between fabric nodes. It carries no client state.

2

Overlay data plane

VXLAN. The VNI replaces the VLAN as the segment identifier, lifting the ceiling from 4,094 to 16 million.

3

Control plane

LISP. The map server and map resolver answer 'where is this endpoint right now'.

4

Policy plane

TrustSec. An SGT is applied at ingress and enforced as an SGACL at egress — policy follows identity, not IP.

5

Fabric roles

Control plane node, border node, edge node and the wireless LAN controller.

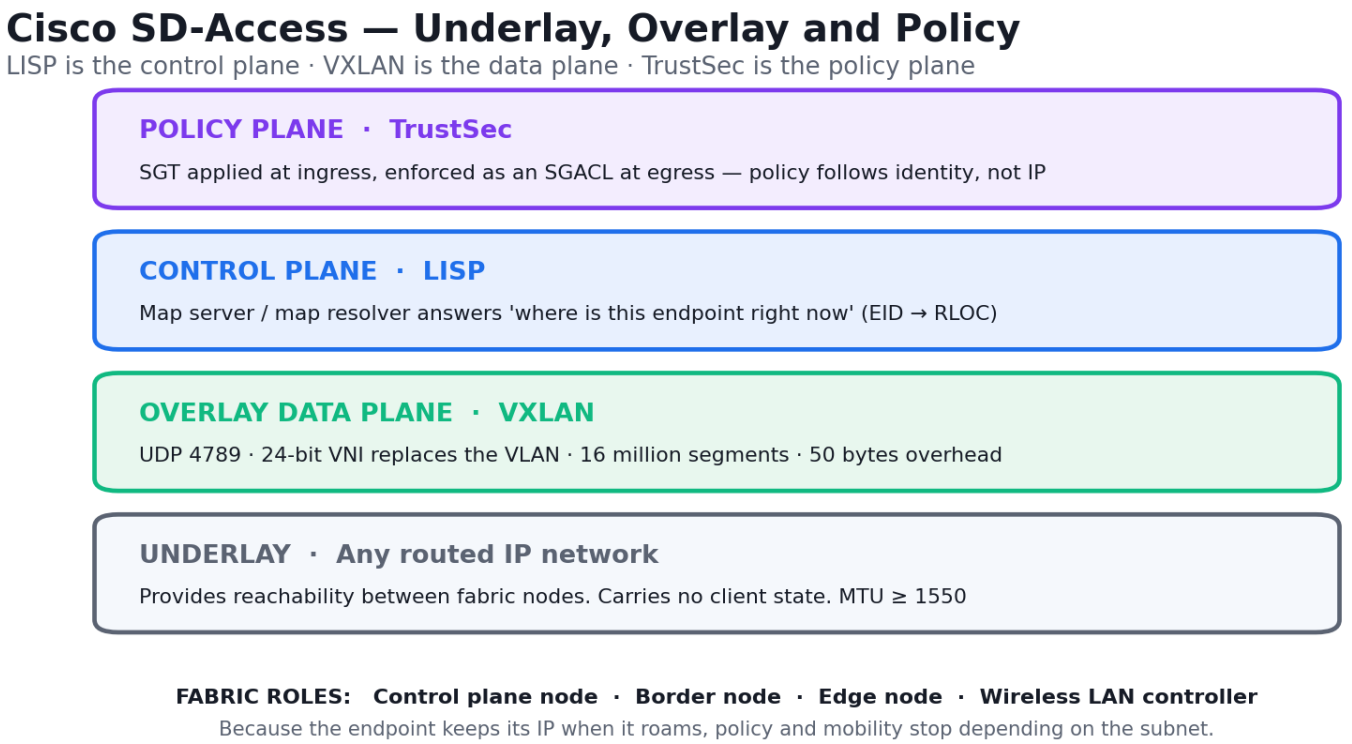
6

Why it matters

Because the endpoint keeps its IP when it roams, policy and mobility both stop depending on the subnet.

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Cisco SD-Access — The Full Stack



LISP resolves where an endpoint is, VXLAN carries the frame there, and TrustSec decides whether it is allowed.

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ENCOR 1.3 AND 1.4

SD-WAN and SD-Access — Do Not Confuse Them

	Cisco SD-WAN	Cisco SD-Access
Where it applies	The WAN, between sites	The campus, inside a site
Control plane	OMP via vSmart	LISP via the control plane node
Data plane	IPsec tunnels	VXLAN
Policy	Centralised and localised policy in vManage	SGTs enforced by TrustSec
Managed by	vManage	Cisco DNA Center

WHEN IT MATTERS Both are 'software-defined' and both appear in domain 1, but they solve different problems in different places. Exam questions rely on you keeping them apart.

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01

TOPIC 01

Enterprise Architecture

Design principles · wireless design · SD-WAN · SD-Access · QoS · switching mechanisms

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Topic 01 — What This Domain Covers

1

Exam domain

ENCOR domain 1.0 Architecture

2

Exam weighting

15% of the ENCOR 350-401 exam

3

TSC mapping

Skills Framework K1, A1 · Network Configuration (ICT-DIT-3009-1.1)

4

Hands-on labs

Labs 1–5 — every objective has a lab

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TOPIC WEIGHTING 15%

Key Concepts — Enterprise Architecture

1. Enterprise campus designs are layered: a 2-tier (collapsed core) design merges core and distribution and suits a single building, while a 3-tier design keeps a dedicated core and is justified once you exceed roughly three distribution blocks.

2. The access layer terminates end devices and enforces port security, PoE and QoS trust; the distribution layer aggregates and is the boundary where Layer 2 becomes Layer 3; the core is a high-speed, low-latency transit fabric that must never run policy.

3. High availability is delivered in three planes: redundancy of hardware and links, FHRP (HSRP, VRRP, GLBP) for default-gateway resilience, and SSO/NSF so a supervisor failover does not tear down routing adjacencies.

4. Wireless deployment models trade control against latency — centralized (all traffic tunnelled to a WLC), distributed/FlexConnect (locally switched at the branch), controller-less (Mobility Express), cloud-managed (Meraki) and remote branch.

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TOPIC WEIGHTING 15%

Key Concepts — Enterprise Architecture (2)

1. Location services (RSSI trilateration, Angle of Arrival with Hyperlocation, BLE beacons) drive asset tracking and analytics; accuracy depends on AP density and placement, not on client radio power.

2. Client density planning is a capacity problem, not a coverage problem: high-density areas need smaller cells, lower AP transmit power, narrower channel widths and band steering to 5 GHz.

3. Cisco SD-WAN separates a control plane (vSmart), an orchestration plane (vBond), a management plane (vManage) and a data plane (vEdge/cEdge) — OMP distributes routes and policy, and IPsec builds the fabric.

4. Cisco SD-Access builds a VXLAN overlay on any IP underlay: the control plane is LISP (map server / map resolver), the data plane is VXLAN, and policy is carried as SGTs enforced by TrustSec.

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TOPIC WEIGHTING 15%

Key Concepts — Enterprise Architecture (3)

1. QoS components are classification and marking (DSCP/CoS), queuing and scheduling (LLQ/CBWFQ), congestion avoidance (WRED), policing (drops) and shaping (buffers) — mark once, at the trust boundary, and act everywhere.

2. Wireless QoS maps DSCP to the 802.11e user-priority (WMM) access categories; a mismatch between the wired DSCP and the wireless UP is the most common reason voice quality collapses over Wi-Fi.

3. Hardware switching is a table story: CAM is exact-match (MAC address table), TCAM is ternary and stores masked entries for ACLs and QoS, the FIB is the CEF-prepared copy of the RIB, and the adjacency table holds the rewrite information.

4. Cisco Express Forwarding replaces process and fast switching by building the FIB and adjacency table ahead of traffic — so the first packet of a flow is switched in hardware just like the millionth.

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WHAT YOU'LL DO

Hands-On Labs — Enterprise Architecture

Labs 1–2

- Investigate the CAM and TCAM Tables
- Analyse Cisco Express Forwarding

Labs 3–4

- Compare Two-Tier and Three-Tier Campus Designs
- Interpret Wired and Wireless QoS Configurations

Lab 5

- Explore SD-WAN and SD-Access Architecture

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TOPIC 01 · HANDS-ON

Investigate the CAM and TCAM Tables

LAB 1

Describe hardware switching mechanisms such as CAM, TCAM, FIB, RIB and adjacency tables (ENCOR 1.6).

You take the role of a campus network engineer verifying how an access switch actually forwards a frame. You generate traffic between hosts, watch the MAC address table (CAM) populate by learning source addresses, then age an entry out and observe the switch flood the unknown unicast. You finish by inspecting how TCAM stores the masked ACL and QoS entries that CAM's exact-match lookup cannot hold.

YOU'LL BUILD

A documented CAM table showing dynamically learned entries, one static entry you added, and a written comparison of CAM versus TCAM lookup behaviour.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, S1/S2 access switches, PC-A and PC-B

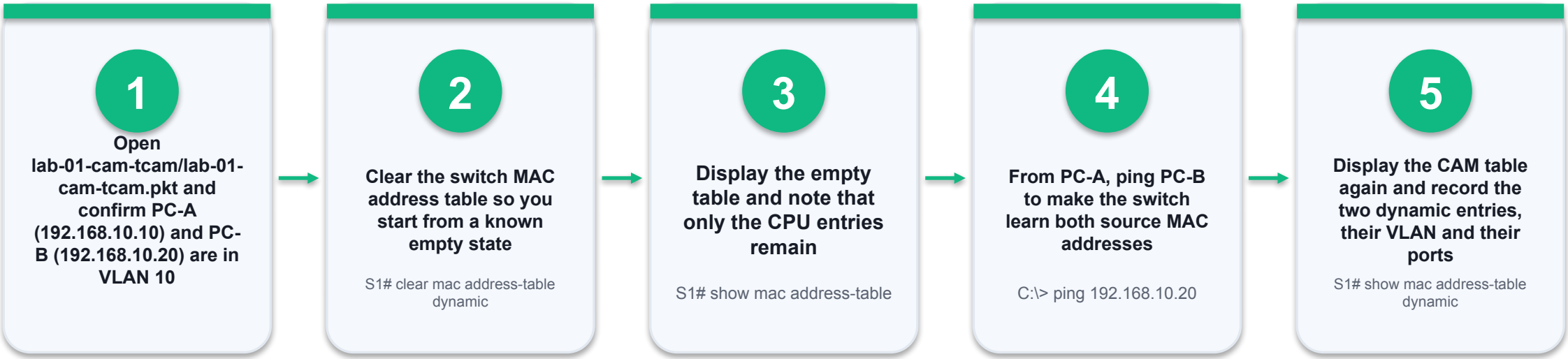
DONE WHEN

'show mac address-table dynamic' lists both PC MAC addresses against the correct ports and VLAN 10, 'show mac address-table static' shows your pinned entry, and you can state that CAM does exact-match on MAC while TCAM does masked match for ACL/QoS.

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LAB 1 · PROCESS MAP

How Lab 1 Runs



THE POINT

Describe hardware switching mechanisms such as CAM, TCAM, FIB, RIB and adjacency tables (ENCOR 1.6).

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LAB 1 · VERIFY

Investigate the CAM and TCAM Tables

✅ Expected result

'show mac address-table dynamic' lists both PC MAC addresses against the correct ports and VLAN 10, 'show mac address-table static' shows your pinned entry, and you can state that CAM does exact-match on MAC while TCAM does masked match for ACL/QoS.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 01 · HANDS-ON

Analyse Cisco Express Forwarding

LAB 2

Describe and verify CEF, the FIB and the adjacency table, and contrast them with process and fast switching (ENCOR 1.6).

Continuing as the campus engineer, you now move up to Layer 3 and prove how the router forwards in hardware. You examine the RIB, then the FIB that CEF builds from it ahead of any traffic, then the adjacency table that holds the Layer 2 rewrite. You disable CEF to watch forwarding fall back to process switching, observe the CPU cost, and re-enable it.

YOU'LL BUILD

A side-by-side record of the RIB, the FIB and the adjacency table for one destination prefix, plus measured evidence of the CPU difference when CEF is disabled.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1 and R2 routers, PC-A and PC-B

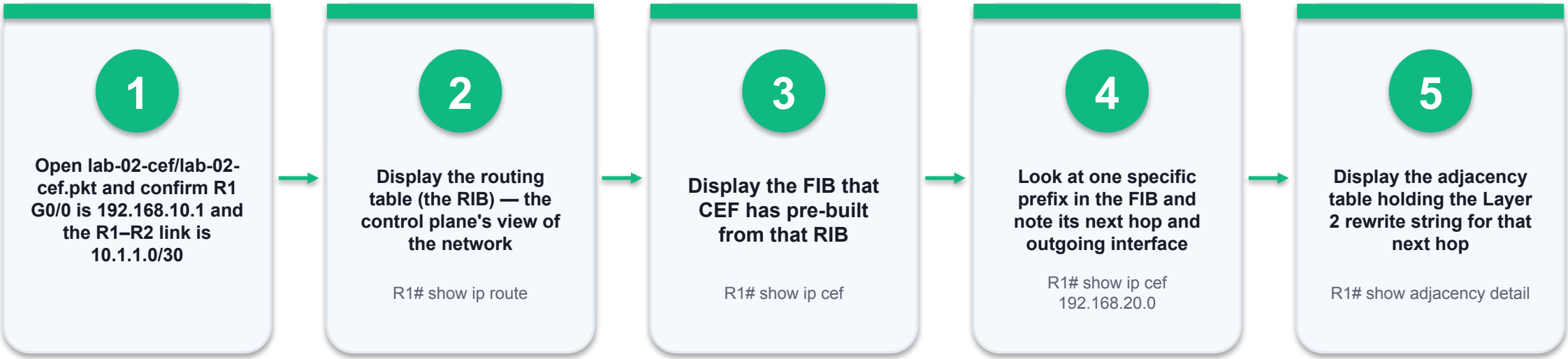
DONE WHEN

'show ip cef' returns a populated FIB and 'show adjacency detail' shows a valid rewrite for the next hop; with CEF disabled you can point to a measurably higher CPU figure during the ping, and it falls again once CEF is re-enabled.

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LAB 2 · PROCESS MAP

How Lab 2 Runs



THE POINT

Describe and verify CEF, the FIB and the adjacency table, and contrast them with process and fast switching (ENCOR 1.6).

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LAB 2 · VERIFY

Analyse Cisco Express Forwarding

✅ Expected result

'show ip cef' returns a populated FIB and 'show adjacency detail' shows a valid rewrite for the next hop; with CEF disabled you can point to a measurably higher CPU figure during the ping, and it falls again once CEF is re-enabled.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

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ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 01 · HANDS-ON

Compare Two-Tier and Three-Tier Campus Designs

LAB 3

Explain enterprise design principles including 2-tier, 3-tier, fabric and cloud designs (ENCOR 1.1.a).

A 400-staff company on one campus is merging with a 2,000-staff company across four buildings. As the design engineer you must recommend a campus architecture for the merged organisation. You document the current 2-tier collapsed-core design, model the merged requirement, and justify whether a dedicated core layer is now warranted — the exact judgement the exam tests.

YOU'LL BUILD

A completed design comparison table and a one-page written recommendation stating whether to move to a 3-tier design, with the trigger conditions that justify it.

TOOLCHAIN

Cisco Packet Tracer, network diagram, design decision worksheet

DONE WHEN

Your recommendation names 3-tier for the merged network, shows the full-mesh link count that falls from six inter-block links to four uplinks once a core is added, and states that the core layer carries no ACLs or policy.

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LAB 3 · PROCESS MAP

How Lab 3 Runs

1. **Open lab-03-campus-design/lab-03-campus-design.pkt and identify which switches are access and which are distribution**
2. **Record what each layer is doing today — port security and PoE at access, routing at distribution**

   SW-DIST-1# show ip route
3. **Confirm the distribution layer is the Layer 2 to Layer 3 boundary by checking the SVIs**

   SW-DIST-1# show ip interface brief | include Vlan
4. **Count the distribution blocks the merged network will need (one per building)**
5. **Apply the design rule: a dedicated core is justified beyond roughly three distribution blocks**

THE POINT

Explain enterprise design principles including 2-tier, 3-tier, fabric and cloud designs (ENCOR 1.1.a).

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LAB 3 · VERIFY

Compare Two-Tier and Three-Tier Campus Designs

✅ Expected result

Your recommendation names 3-tier for the merged network, shows the full-mesh link count that falls from six inter-block links to four uplinks once a core is added, and states that the core layer carries no ACLs or policy.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 01 · HANDS-ON

Interpret Wired and Wireless QoS Configurations

LAB 4

Interpret wired and wireless QoS configurations, including QoS components and QoS policy (ENCOR 1.5).

Voice calls on the campus wireless network are breaking up during busy periods. As the network engineer you trace the QoS treatment of a voice packet end to end: you read the classification and marking at the wired trust boundary, examine the queuing policy, then check how DSCP maps to the 802.11e WMM access category on the wireless side — where the mismatch usually is.

YOU'LL BUILD

A documented end-to-end QoS policy showing the class map, policy map and service policy, plus the DSCP-to-WMM mapping table that explains the voice degradation.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1 router, SW-1 switch, WLC

DONE WHEN

You can trace a voice packet from the phone's EF marking, through the trusted access port, into the priority queue on the WAN interface, and state which DSCP-to-WMM mapping is wrong and why calls break up.

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LAB 4 · PROCESS MAP

How Lab 4 Runs



THE POINT

Interpret wired and wireless QoS configurations, including QoS components and QoS policy (ENCOR 1.5).

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LAB 4 · VERIFY

Interpret Wired and Wireless QoS Configurations

✅ Expected result

You can trace a voice packet from the phone's EF marking, through the trusted access port, into the priority queue on the WAN interface, and state which DSCP-to-WMM mapping is wrong and why calls break up.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 01 · HANDS-ON

Explore SD-WAN and SD-Access Architecture

LAB 5

Explain the working principles of the Cisco SD-WAN and SD-Access solutions and their control and data plane elements (ENCOR 1.3, 1.4).

Your organisation is evaluating both Cisco SD-WAN for its branch WAN and SD-Access for its campus. As the architect you must be able to name every plane element and say what breaks if it fails. You map the four SD-WAN planes and the SD-Access LISP/VXLAN/TrustSec stack, then work through failure scenarios that the exam poses as 'what happens if…' questions.

YOU'LL BUILD

Two completed architecture maps — SD-WAN's four planes and SD-Access's control/data/policy planes — plus a failure-impact table for each component.

TOOLCHAIN

Cisco Packet Tracer, architecture worksheet, sdwan-components.csv, sdaccess-components.csv

DONE WHEN

You can name all four SD-WAN planes with their controller, state that a vSmart outage stops new route learning but not existing forwarding, and correctly identify LISP as the SD-Access control plane and VXLAN as its data plane.

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LAB 5 · PROCESS MAP

How Lab 5 Runs



THE POINT

Explain the working principles of the Cisco SD-WAN and SD-Access solutions and their control and data plane elements (ENCOR 1.3, 1.4).

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LAB 5 · VERIFY

Explore SD-WAN and SD-Access Architecture

✅ Expected result

You can name all four SD-WAN planes with their controller, state that a vSmart outage stops new route learning but not existing forwarding, and correctly identify LISP as the SD-Access control plane and VXLAN as its data plane.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC RECAP

Recap — Enterprise Architecture

- You can now: Describe hardware switching mechanisms such as CAM, TCAM, FIB, RIB and adjacency tables (ENCOR 1.6).
- You can now: Describe and verify CEF, the FIB and the adjacency table, and contrast them with process and fast switching (ENCOR 1.6).
- You can now: Explain enterprise design principles including 2-tier, 3-tier, fabric and cloud designs (ENCOR 1.1.a).
- You can now: Interpret wired and wireless QoS configurations, including QoS components and QoS policy (ENCOR 1.5).
- You can now: Explain the working principles of the Cisco SD-WAN and SD-Access solutions and their control and data plane elements (ENCOR 1.3, 1.4).

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02

TOPIC 02

Virtualization Technologies

Hypervisors · virtual machines · virtual switching · VRF · GRE and IPsec · LISP and VXLAN

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Topic 02 — What This Domain Covers

1

Exam domain

ENCOR domain 2.0 Virtualization

2

Exam weighting

10% of the ENCOR 350-401 exam

3

TSC mapping

Skills Framework K2, A2 · Network Configuration (ICT-DIT-3009-1.1)

4

Hands-on labs

Labs 6–9 — every objective has a lab

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TOPIC WEIGHTING 10%

Key Concepts — Virtualization Technologies

1. A Type 1 (bare-metal) hypervisor such as ESXi, KVM or Hyper-V runs directly on the hardware and is what data centres deploy; a Type 2 hypervisor such as VMware Workstation or VirtualBox runs as an application on a host OS and suits a lab.

2. A virtual machine packages vCPU, vRAM, vNIC and virtual disk with a full guest OS — heavier than a container but fully isolated, which is why network functions (CSR 1000v, ASAv, WLC-v) ship as VMs.

3. Virtual switching (vSwitch, Cisco Nexus 1000v, DVS) moves the access layer inside the hypervisor: VM-to-VM traffic on the same host never touches a physical switch, which is exactly why it can bypass your physical ACLs.

4. VRF (Virtual Routing and Forwarding) creates multiple independent routing tables on one device — the same IP prefix can exist in two VRFs without conflict, giving Layer 3 segmentation without extra hardware.

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TOPIC WEIGHTING 10%

Key Concepts — Virtualization Technologies (2)

1. VRF-Lite is VRF without MPLS: you define the VRF, assign interfaces with 'vrf forwarding', and every show/ping/traceroute must then be qualified with 'vrf <name>' or it consults the global table and appears to fail.

2. GRE tunnels are simple, multiprotocol and support routing protocols and multicast — but they provide no confidentiality, integrity or authentication whatsoever.

3. IPsec provides confidentiality, integrity, authentication and anti-replay; the classic enterprise pattern is GRE over IPsec, which gives you the routing and multicast support of GRE plus the protection of IPsec.

4. IPsec negotiates in two phases: IKE Phase 1 builds a bidirectional management tunnel (ISAKMP SA), and Phase 2 builds the unidirectional IPsec SAs that actually carry data — an interesting-traffic ACL mismatch is the classic Phase 2 failure.

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TOPIC WEIGHTING 10%

Key Concepts — Virtualization Technologies (3)

1

LISP splits location from identity: an EID identifies the endpoint and an RLOC identifies where it currently attaches, with a map server / map resolver providing the lookup — this is what makes SD-Access mobility work.

2

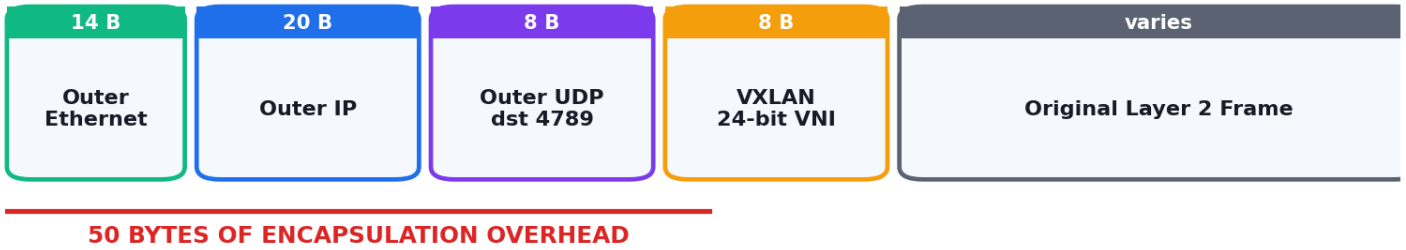
VXLAN encapsulates a Layer 2 frame in UDP (port 4789) and carries it across a routed underlay, extending Layer 2 without spanning tree and lifting the segment ceiling from 4,094 VLANs to 16 million VNIs.

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Decoding the VXLAN Header

The VXLAN Header — 50 Bytes of Overhead

Which is why the underlay MTU must be at least 1550



- **UDP port 4789** — Identifies VXLAN. The outer source port is a hash of the inner flow, which is what enables ECMP across the underlay.
- **24-bit VNI** — 16,777,216 segments, against 4,094 for 802.1Q VLANs — the segment ceiling problem solved.
- **MTU ≥ 1550** — 1500 payload + 50 overhead. An underlay left at 1500 silently fragments or drops.
- **No spanning tree** — Layer 2 is carried over a routed underlay, so STP does not extend across the fabric.

Fifty bytes of overhead is why the underlay MTU must be at least 1550 — an underlay left at 1500 silently drops or fragments.

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WHAT YOU'LL DO

Hands-On Labs — Virtualization Technologies

Labs 6–7

- Configure and Verify VRF-Lite
- Configure and Verify a GRE Tunnel

Labs 8–9

- Configure and Verify a Site-to-Site IPsec VPN
- Explore LISP and VXLAN Network Virtualization

—

- —

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TOPIC 02 · HANDS-ON

Configure and Verify VRF-Lite

LAB 6

Configure and verify data path virtualization technologies — VRF (ENCOR 2.2.a).

Your organisation must keep guest traffic and corporate traffic completely separate on the same physical router, and the acquisition it just completed uses the same 192.168.1.0/24 range as head office. As the network engineer you solve both problems with VRF-Lite: two independent routing tables on one device, with overlapping address space and no leaking between them.

YOU'LL BUILD

A router carrying two VRFs (CORP and GUEST) with overlapping 192.168.1.0/24 prefixes, proven isolated by a VRF-aware ping that succeeds inside a VRF and fails across VRFs.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1 router, CORP-PC and GUEST-PC

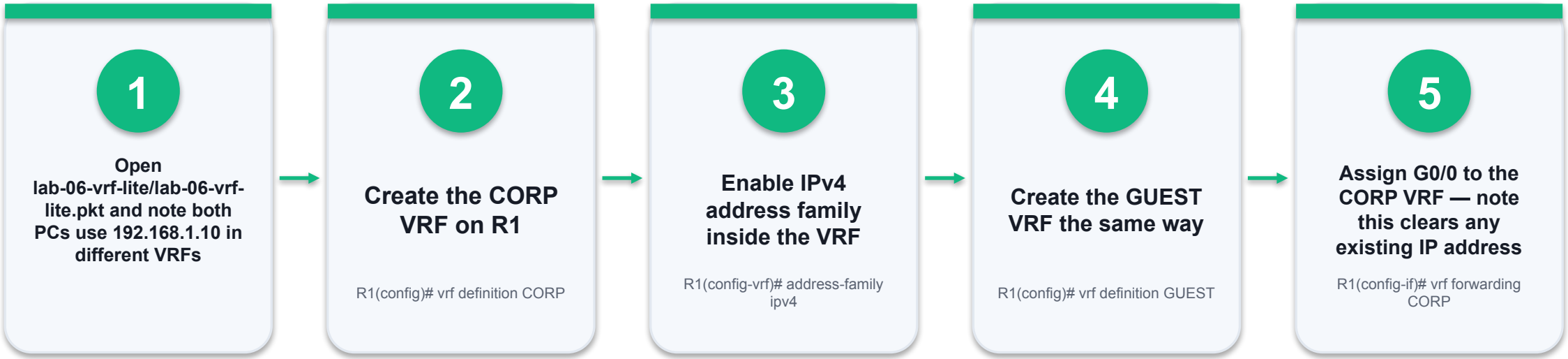
DONE WHEN

'show vrf' lists CORP and GUEST with their interfaces, both VRF routing tables carry 192.168.1.0/24 independently, 'ping vrf CORP' succeeds, and the unqualified ping fails — proving isolation rather than a broken configuration.

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LAB 6 · PROCESS MAP

How Lab 6 Runs



THE POINT

Configure and verify data path virtualization technologies — VRF (ENCOR 2.2.a).

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LAB 6 · VERIFY

Configure and Verify VRF-Lite

✅ Expected result

'show vrf' lists CORP and GUEST with their interfaces, both VRF routing tables carry 192.168.1.0/24 independently, 'ping vrf CORP' succeeds, and the unqualified ping fails — proving isolation rather than a broken configuration.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 02 · HANDS-ON

Configure and Verify a GRE Tunnel

LAB 7

Configure and verify data path virtualization technologies — GRE tunneling (ENCOR 2.2.b).

Head office and the branch are connected only across the public internet, but the routing protocol and multicast traffic they need to exchange cannot cross it. As the WAN engineer you build a GRE tunnel that carries those private networks over the public path, then demonstrate the critical limitation the exam always tests: GRE gives you transport, not confidentiality.

YOU'LL BUILD

A working GRE tunnel between RA (64.103.211.2) and RB (209.165.122.2) carrying 192.168.1.0/24 and 192.168.2.0/24, verified up/up with traffic passing.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, RA and RB routers, PCA and PCB

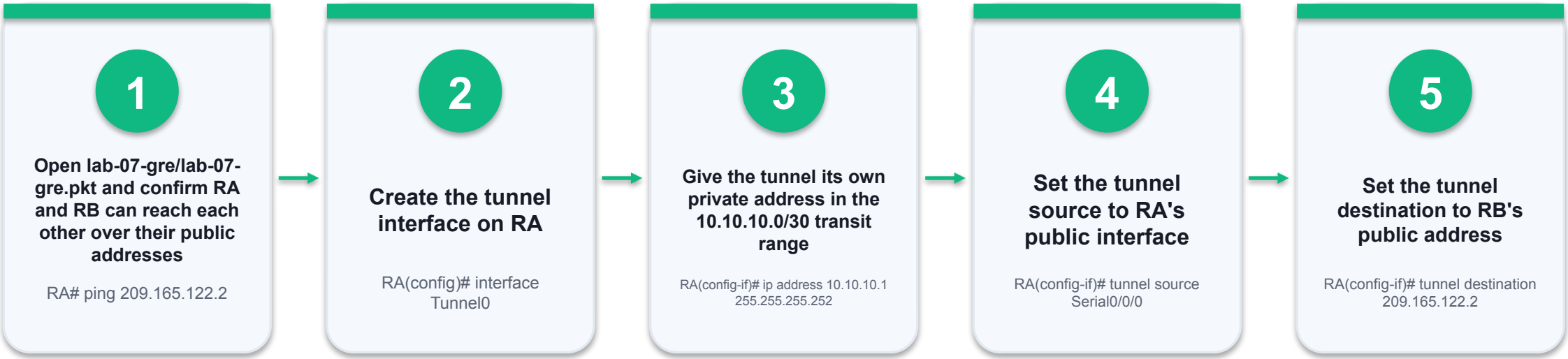
DONE WHEN

'show interfaces Tunnel0' reports Tunnel0 up, line protocol up with the correct source and destination, PCA pings PCB successfully, and you can state that GRE adds no encryption, authentication or integrity.

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LAB 7 · PROCESS MAP

How Lab 7 Runs



THE POINT

Configure and verify data path virtualization technologies — GRE tunneling (ENCOR 2.2.b).

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LAB 7 · VERIFY

Configure and Verify a GRE Tunnel

✅ Expected result

'show interfaces Tunnel0' reports Tunnel0 up, line protocol up with the correct source and destination, PCA pings PCB successfully, and you can state that GRE adds no encryption, authentication or integrity.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 02 · HANDS-ON

Configure and Verify a Site-to-Site IPsec VPN

LAB 8

Configure and verify data path virtualization technologies — IPsec tunneling (ENCOR 2.2.b).

The GRE tunnel from the previous lab carries payroll data in clear text, and the audit has flagged it. As the security-minded network engineer you now build a site-to-site IPsec VPN between R1 and R3 across the R2 ISP, working through both IKE phases and proving the tunnel only forms once interesting traffic matches on both ends.

YOU'LL BUILD

A functioning site-to-site IPsec VPN between R1 (192.168.1.0/24) and R3 (192.168.3.0/24) with a formed ISAKMP SA, established IPsec SAs and encrypted packet counters incrementing.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1/R2/R3 routers, PC-A and PC-C

DONE WHEN

'show crypto isakmp sa' shows the peer in QM_IDLE and 'show crypto ipsec sa' shows non-zero '#pkts encaps' and '#pkts decaps' counters that increase with each ping — traffic between the two LANs is genuinely encrypted.

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LAB 8 · PROCESS MAP

How Lab 8 Runs



THE POINT

Configure and verify data path virtualization technologies — IPsec tunneling (ENCOR 2.2.b).

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LAB 8 · VERIFY

Configure and Verify a Site-to-Site IPsec VPN

✅ Expected result

'show crypto isakmp sa' shows the peer in QM_IDLE and 'show crypto ipsec sa' shows non-zero '#pkts encaps' and '#pkts decaps' counters that increase with each ping — traffic between the two LANs is genuinely encrypted.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 02 · HANDS-ON

Explore LISP and VXLAN Network Virtualization

LAB 9

Describe network virtualization concepts — LISP and VXLAN (ENCOR 2.3.a, 2.3.b).

Your campus is moving to SD-Access, whose entire mobility story rests on LISP and whose data plane is VXLAN. As the network engineer you must be able to trace a packet through both. You walk the LISP map-request and map-reply exchange, follow an endpoint that roams between edge switches, then decode a VXLAN header field by field and calculate the encapsulation overhead.

YOU'LL BUILD

An annotated LISP registration and lookup walkthrough, a completed VXLAN header decode with all field sizes, and a calculated MTU requirement for the underlay.

TOOLCHAIN

Cisco Packet Tracer, packet-decode worksheet, vxlan-header.csv, lisp-mapping.csv

DONE WHEN

You can state the LISP lookup sequence (register, map-request, map-reply, cache), explain why an EID survives a roam, name UDP 4789 as the VXLAN port, and calculate the 50-byte overhead and the 1550-byte minimum underlay MTU.

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LAB 9 · PROCESS MAP

How Lab 9 Runs

1. Open lab-09-lisp-vxlan/ and read the fabric topology showing edge nodes, border node and control plane node
2. Identify the EID — the endpoint identifier, the host's own IP address that never changes as it roams
3. Identify the RLOC — the routing locator, the address of the edge switch the host is attached to now
4. Trace registration: the edge node registers the EID-to-RLOC binding with the map server
5. Trace lookup: an ingress tunnel router sends a map-request to the map resolver

THE POINT

Describe network virtualization concepts — LISP and VXLAN (ENCOR 2.3.a, 2.3.b).

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LAB 9 · VERIFY

Explore LISP and VXLAN Network Virtualization

✅ Expected result

You can state the LISP lookup sequence (register, map-request, map-reply, cache), explain why an EID survives a roam, name UDP 4789 as the VXLAN port, and calculate the 50-byte overhead and the 1550-byte minimum underlay MTU.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC RECAP

Recap — Virtualization Technologies

- You can now: Configure and verify data path virtualization technologies — VRF (ENCOR 2.2.a).
- You can now: Configure and verify data path virtualization technologies — GRE tunneling (ENCOR 2.2.b).
- You can now: Configure and verify data path virtualization technologies — IPsec tunneling (ENCOR 2.2.b).
- You can now: Describe network virtualization concepts — LISP and VXLAN (ENCOR 2.3.a, 2.3.b).

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03

TOPIC 03

Network Infrastructure

Layer 2 · Layer 3 routing · wireless · IP services — the largest exam domain

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Topic 03 — What This Domain Covers

1

Exam domain

ENCOR domain 3.0 Infrastructure

2

Exam weighting

30% of the ENCOR 350-401 exam

3

TSC mapping

Skills Framework K3, A3 · Network Configuration (ICT-DIT-3009-1.1)

4

Hands-on labs

Labs 10–21 — every objective has a lab

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TOPIC WEIGHTING 30%

Key Concepts — Network Infrastructure

1. An 802.1Q trunk carries multiple VLANs on one link by inserting a 4-byte tag; the native VLAN is sent untagged, and a native-VLAN mismatch between two ends produces VLAN leaking that CDP will warn you about.

2. DTP negotiates trunking automatically, which is convenient and insecure — production practice is to hard-set 'switchport mode trunk' or 'access' and disable negotiation with 'switchport nonegotiate'.

3. EtherChannel bundles up to eight physical links into one logical link; PAgP is Cisco-proprietary (desirable/auto) and LACP is the IEEE 802.3ad standard (active/passive) — two 'auto' or two 'passive' ends will never form a bundle.

4. EtherChannel members must match on speed, duplex, trunk mode, allowed VLAN list and native VLAN — a mismatch on any of these suspends the port, and 'show etherchannel summary' flags it.

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TOPIC WEIGHTING 30%

Key Concepts — Network Infrastructure (2)

1. STP elects a root bridge by lowest bridge ID (priority + MAC), then each non-root switch picks one root port by lowest cumulative path cost, then each segment picks a designated port — every remaining port is blocked.

2. RSTP (802.1w) converges in seconds rather than the 30–50 seconds of legacy STP by using proposal/agreement on point-to-point links, plus alternate and backup port roles that are pre-computed standby paths.

3. MST (802.1s) maps many VLANs onto a few instances so a 1,000-VLAN campus runs two or three spanning trees instead of a thousand — switches only share an MST region when name, revision and VLAN-to-instance map match exactly.

4. STP protection features each defend a different attack: PortFast skips listening/learning on edge ports, BPDU guard shuts a PortFast port that receives a BPDU, root guard prevents a superior BPDU from moving the root, and loop guard covers unidirectional link failures.

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TOPIC WEIGHTING 30%

Key Concepts — Network Infrastructure (3)

1. EIGRP is an advanced distance-vector protocol using DUAL: the successor is the best path, and a feasible successor is a guaranteed loop-free backup that meets the feasibility condition (its reported distance is less than the current feasible distance).

2. OSPF is a link-state protocol: every router floods LSAs, builds an identical LSDB for the area, and independently runs Dijkstra's SPF to compute its own shortest-path tree — cost is the metric, by default reference bandwidth over interface bandwidth.

3. OSPF area types control LSA flooding: a stub area blocks external LSAs (Type 5), a totally stubby area also blocks summaries (Type 3), and NSSA allows local redistribution via Type 7 LSAs that the ABR translates to Type 5.

4. OSPF neighbours only form when hello/dead timers, area ID, subnet mask, authentication and stub flags all match; on broadcast networks a DR and BDR are elected to prevent an n-squared adjacency mesh.

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TOPIC WEIGHTING 30%

Key Concepts — Network Infrastructure (4)

1. eBGP is a path-vector protocol between autonomous systems: it forms TCP 179 sessions with explicitly configured neighbours, and its best-path algorithm evaluates weight, local preference, AS-path length, origin, MED and more — in that order.

2. eBGP peers are normally directly connected and default to a TTL of 1; peering to a loopback therefore requires 'ebgp-multihop' and 'update-source', which is the most common eBGP lab failure.

3. Policy-based routing overrides the routing table for selected traffic using a route-map matched on an ACL, letting you send (say) guest traffic out a different path than the RIB's best path.

4. Wireless Layer 1 is a power budget: RSSI is the received signal strength, SNR is the margin between signal and noise floor, and a healthy data connection wants roughly -67 dBm RSSI and 25 dB SNR.

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TOPIC WEIGHTING 30%

Key Concepts — Network Infrastructure (5)

1. AP modes serve different jobs — local (normal service), FlexConnect (branch local switching), monitor (scanning only), sniffer (packet capture), bridge/mesh, and SE-Connect for spectrum analysis.

2. An AP joins a WLC by CAPWAP: it discovers candidate controllers (broadcast, DHCP option 43, DNS, or a previously learnt list), selects one by priority and AP capacity, then builds a DTLS-encrypted CAPWAP control tunnel on UDP 5246 and a data tunnel on UDP 5247.

3. Layer 2 roaming keeps the client in the same subnet so only the AP association changes; Layer 3 roaming crosses a subnet boundary and requires an anchor/foreign controller tunnel so the client's IP address survives the move.

4. NTP disciplines clocks in a stratum hierarchy so logs, certificates and troubleshooting timelines correlate; PTP (1588) provides sub-microsecond hardware-timestamped accuracy where NTP's millisecond accuracy is not enough.

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TOPIC WEIGHTING 30%

Key Concepts — Network Infrastructure (6)

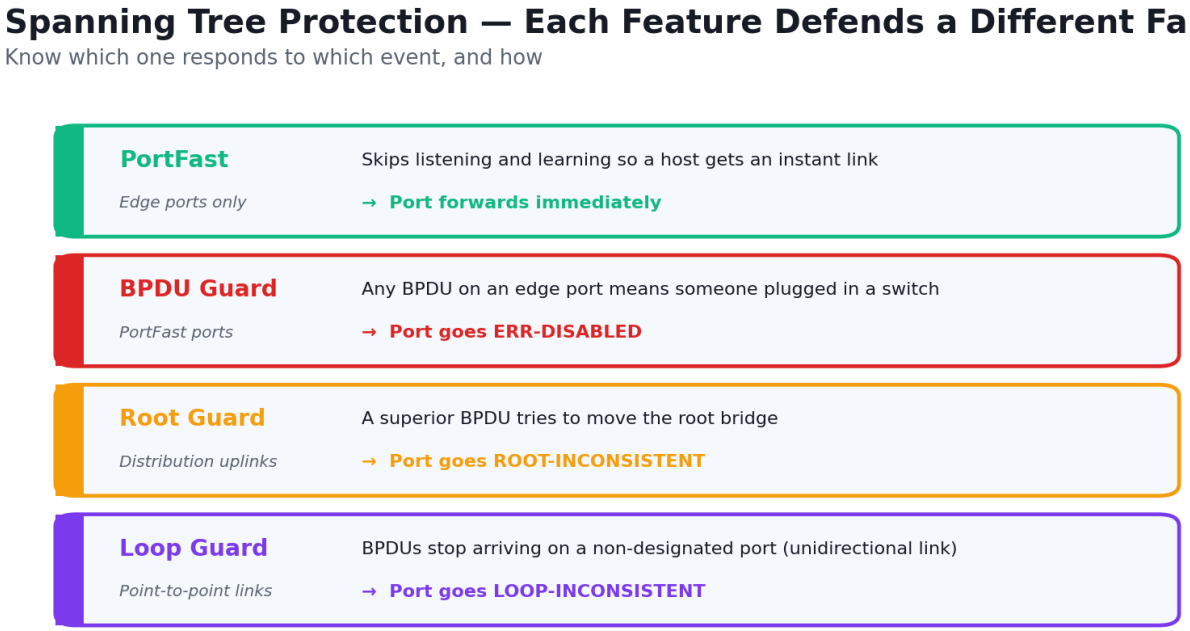
1. NAT translates a private inside-local address to a public inside-global address; PAT overloads one public address across many hosts by also translating the source port, which is why 'ip nat inside source list 1 interface g0/0 overload' is the ubiquitous branch configuration.

2. First-hop redundancy protocols present one virtual IP and virtual MAC to hosts: HSRP is Cisco (active/standby, group 0 default, preemption off by default), VRRP is the standard (master/backup, preemption on by default), and GLBP additionally load-balances across gateways.

3. Multicast delivers one stream to many receivers: IGMP is how hosts join a group on the LAN, PIM builds the distribution tree between routers, and the Reverse Path Forwarding check discards any multicast packet that did not arrive on the interface the router would use to reach the source.

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Spanning Tree Protection Features



Exam point: BPDU guard ERR-DISABLES the port (it must be recovered). Root guard puts it in root-inconsistent, which clears by itself once the superior BPDU stops. That difference is examined directly.

BPDU guard err-disables the port and must be recovered; root guard's root-inconsistent state clears by itself. That difference is examined.

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WHAT YOU'LL DO

Hands-On Labs — Network Infrastructure

Labs 10–13

- Troubleshoot 802.1Q Trunking and DTP
- Troubleshoot Static and Dynamic EtherChannels
- Configure RSTP and Tune the Spanning Tree
- Configure Multiple Spanning Tree (MST)

Labs 14–17

- Harden the Spanning Tree with PortFast, BPDU Guard and Root Guard
- Compare EIGRP and OSPF Path Selection
- Configure Multiarea OSPFv2 with Summarization and Filtering
- Configure Multiarea OSPFv3 for IPv6

Labs 18–21

- Configure and Verify eBGP
- Configure a Basic WLAN and WPA2-Enterprise on the WLC
- Troubleshoot WLAN and Wireless Client Connectivity
- Configure NTP, NAT/PAT and HSRP

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TOPIC 03 · HANDS-ON

Troubleshoot 802.1Q Trunking and DTP

LAB 10

Troubleshoot static and dynamic 802.1q trunking protocols (ENCOR 3.1.a).

Users in VLAN 20 on the second floor have lost access to their file server while VLAN 10 users are unaffected. As the support engineer you work the trunk between S1 and S2: you verify the trunk formed at all, check the allowed VLAN list that is silently pruning VLAN 20, and resolve a native VLAN mismatch that CDP is already warning you about in the logs.

YOU'LL BUILD

A repaired 802.1Q trunk carrying VLANs 10, 20 and 99 with a matching native VLAN on both ends, and DTP disabled in favour of a hard-set trunk.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, S1/S2/S3 switches, PC0–PC5

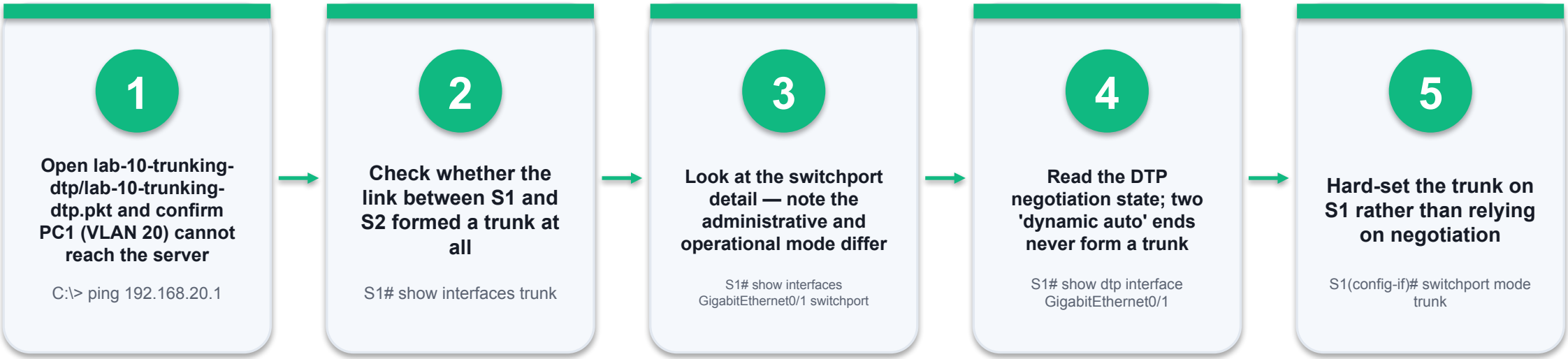
DONE WHEN

'show interfaces trunk' reports mode 'on' with VLANs 10, 20 and 99 allowed and native VLAN 99 on both switches, the mismatch log stops, and the VLAN 20 PC pings its server.

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LAB 10 · PROCESS MAP

How Lab 10 Runs



THE POINT

Troubleshoot static and dynamic 802.1q trunking protocols (ENCOR 3.1.a).

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LAB 10 · VERIFY

Troubleshoot 802.1Q Trunking and DTP

✅ Expected result

'show interfaces trunk' reports mode 'on' with VLANs 10, 20 and 99 allowed and native VLAN 99 on both switches, the mismatch log stops, and the VLAN 20 PC pings its server.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 03 · HANDS-ON

Troubleshoot Static and Dynamic EtherChannels

LAB 11

Troubleshoot static and dynamic EtherChannels — PAgP and LACP (ENCOR 3.1.b).

Three EtherChannel bundles were configured across the distribution layer over the weekend and only one came up. As the engineer on duty you diagnose each: one has both ends passive so LACP never initiates, one has a speed and duplex mismatch that suspends a member port, and one mixes PAgP on one end with LACP on the other — a combination that can never negotiate.

YOU'LL BUILD

Three working EtherChannel bundles — Po1 with PAgP, Po2 with LACP and Po3 negotiated LACP — all showing SU (in use) with every member port bundled.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, S1/S2/S3 switches

DONE WHEN

'show etherchannel summary' shows Po1, Po2 and Po3 all with flags (SU) and every member port marked (P) for bundled — no (s) suspended and no (D) down ports remain.

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LAB 11 · PROCESS MAP

How Lab 11 Runs



THE POINT

Troubleshoot static and dynamic EtherChannels — PAgP and LACP (ENCOR 3.1.b).

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LAB 11 · VERIFY

Troubleshoot Static and Dynamic EtherChannels

✅ Expected result

'show etherchannel summary' shows Po1, Po2 and Po3 all with flags (SU) and every member port marked (P) for bundled — no (s) suspended and no (D) down ports remain.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 03 · HANDS-ON

Configure RSTP and Tune the Spanning Tree

LAB 12

Configure and verify common Spanning Tree Protocols — RSTP — and Spanning Tree enhancements (ENCOR 3.1.c).

The spanning tree in the distribution block elected an access switch as root, so traffic between two distribution switches is taking a suboptimal path through the wiring closet. As the engineer you deterministically move the root where it belongs, tune the port costs, migrate the network from legacy PVST+ to Rapid PVST+, and measure the convergence improvement.

YOU'LL BUILD

A spanning tree with S1 as the deterministic root bridge and S2 as the secondary, running Rapid PVST+, with a measured convergence time on link failure.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, S1/S2/S3 switches, PC-A

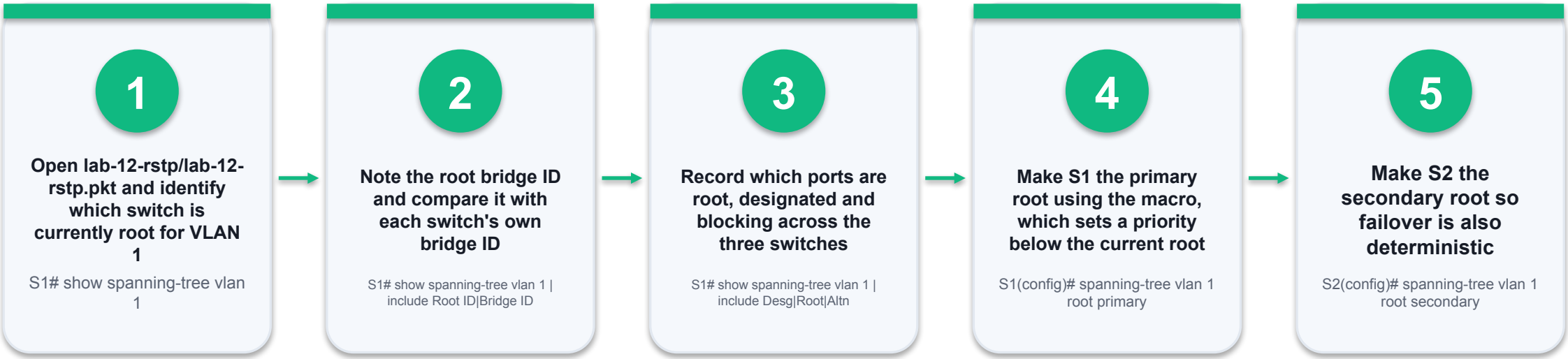
DONE WHEN

'show spanning-tree vlan 1' on S1 states 'This bridge is the root' with priority 4096, all switches report protocol rstp, and your recorded convergence after the link failure is seconds rather than the 30–50 seconds of legacy STP.

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LAB 12 · PROCESS MAP

How Lab 12 Runs



THE POINT

Configure and verify common Spanning Tree Protocols — RSTP — and Spanning Tree enhancements (ENCOR 3.1.c).

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LAB 12 · VERIFY

Configure RSTP and Tune the Spanning Tree

✅ Expected result

'show spanning-tree vlan 1' on S1 states 'This bridge is the root' with priority 4096, all switches report protocol rstp, and your recorded convergence after the link failure is seconds rather than the 30–50 seconds of legacy STP.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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Configure Multiple Spanning Tree (MST)

LAB 13

Configure and verify common Spanning Tree Protocols — MST (ENCOR 3.1.c).

The campus carries 200 VLANs, and running one spanning tree per VLAN is consuming switch CPU and making troubleshooting impractical. As the engineer you consolidate them onto two MST instances, load-balancing the VLANs across two roots, and confirm all switches join the same MST region — which requires name, revision and VLAN map to match exactly.

YOU'LL BUILD

A three-switch MST region named CAMPUS with VLANs mapped to instances 1 and 2, S1 root for instance 1 and S2 root for instance 2, giving true load distribution.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, S1/S2/S3 switches

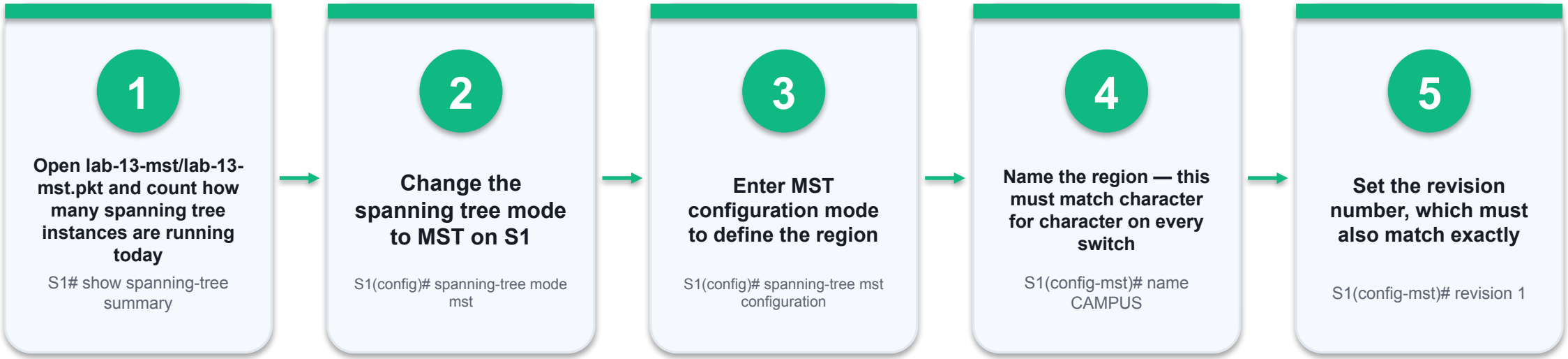
DONE WHEN

'show spanning-tree mst configuration' returns the same region name CAMPUS, revision 1 and identical configuration digest on all three switches, and 'show spanning-tree mst' shows S1 root for instance 1 and S2 root for instance 2.

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LAB 13 · PROCESS MAP

How Lab 13 Runs



THE POINT

Configure and verify common Spanning Tree Protocols — MST (ENCOR 3.1.c).

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LAB 13 · VERIFY

Configure Multiple Spanning Tree (MST)

✅ Expected result

'show spanning-tree mst configuration' returns the same region name CAMPUS, revision 1 and identical configuration digest on all three switches, and 'show spanning-tree mst' shows S1 root for instance 1 and S2 root for instance 2.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 03 · HANDS-ON

Harden the Spanning Tree with PortFast, BPDU Guard and Root Guard

LAB 14

Configure and verify Spanning Tree enhancements such as root guard and BPDU guard (ENCOR 3.1.c).

A contractor plugged a personal switch into a meeting-room wall port last month and became the spanning tree root, black-holing the floor for four minutes. As the engineer you make that impossible: PortFast on edge ports so hosts get an instant link, BPDU guard to shut any edge port that receives a BPDU, and root guard on the distribution uplinks.

YOU'LL BUILD

Edge ports running PortFast with BPDU guard globally enabled, root guard on the distribution-facing uplinks, and a demonstrated err-disable when a rogue BPDU arrives.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, S1/S2/S3 switches, rogue switch, PC-A

DONE WHEN

The rogue switch's BPDU puts the PortFast port into err-disabled with a BPDUGUARD log entry, 'show spanning-tree inconsistentports' reports a root-inconsistent port when a superior BPDU hits the guarded uplink, and errdisable recovery is armed.

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LAB 14 · PROCESS MAP

How Lab 14 Runs

1. **Open lab-14-stp-hardening/lab-14-stp-hardening.pkt and time how long a PC port takes to forward**
2. **Enable PortFast on the access port so it skips listening and learning**
   S1(config-if)# spanning-tree portfast
3. **Enable PortFast by default on all non-trunking ports**
   S1(config)# spanning-tree portfast default
4. **Verify PortFast is active on the interface**
   S1# show spanning-tree interface FastEthernet0/1 portfast
5. **Enable BPDU guard on that specific interface**
   S1(config-if)# spanning-tree bpduguard enable

THE POINT

Configure and verify Spanning Tree enhancements such as root guard and BPDU guard (ENCOR 3.1.c).

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LAB 14 · VERIFY

Harden the Spanning Tree with PortFast, BPDU Guard and Root Guard

✅ Expected result

The rogue switch's BPDU puts the PortFast port into err-disabled with a BPDUGUARD log entry, 'show spanning-tree inconsistentports' reports a root-inconsistent port when a superior BPDU hits the guarded uplink, and errdisable recovery is armed.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 03 · HANDS-ON

Compare EIGRP and OSPF Path Selection

LAB 15

Compare routing concepts of EIGRP and OSPF — advanced distance vector versus link state, load balancing, path selection, metrics and area types (ENCOR 3.2.a).

Your network runs EIGRP in the legacy branch estate and OSPF in the new campus, and you must explain to management why the two choose different paths across the identical topology. As the engineer you configure both over the same links, compare their metrics, and demonstrate the one capability EIGRP has that OSPF does not: unequal-cost load balancing.

YOU'LL BUILD

Both protocols running over one topology with documented metric calculations, an identified successor and feasible successor in EIGRP, and working unequal-cost load balancing via the variance command.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1/R2/R3 routers

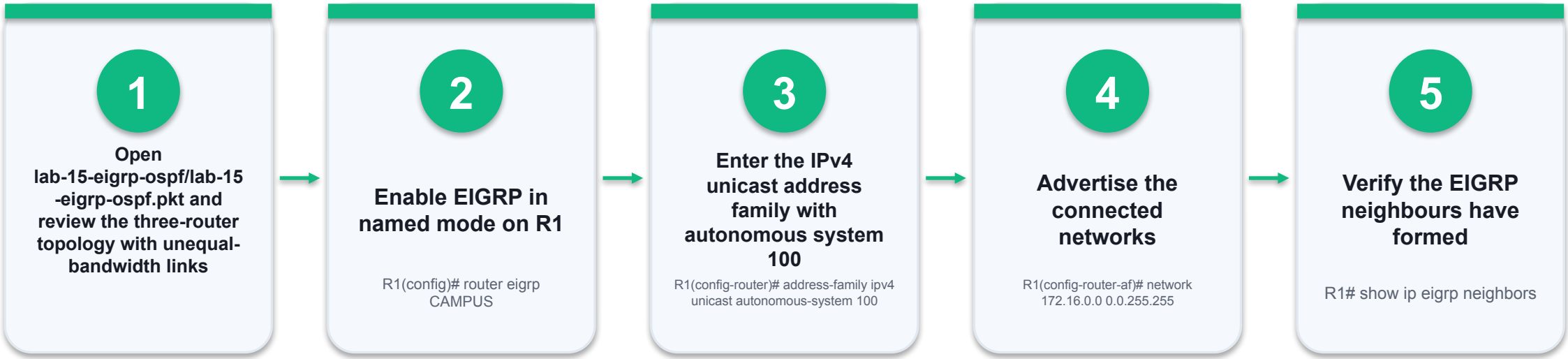
DONE WHEN

'show ip eigrp topology' identifies a successor and at least one feasible successor with its reported distance, the routing table shows two unequal-cost paths after variance 2, and OSPF neighbours reach FULL with a cost you can calculate from reference bandwidth over interface bandwidth.

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LAB 15 · PROCESS MAP

How Lab 15 Runs



THE POINT

Compare routing concepts of EIGRP and OSPF — advanced distance vector versus link state, load balancing, path selection, metrics and area types (ENCOR 3.2.a).

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LAB 15 · VERIFY

Compare EIGRP and OSPF Path Selection

✅ Expected result

'show ip eigrp topology' identifies a successor and at least one feasible successor with its reported distance, the routing table shows two unequal-cost paths after variance 2, and OSPF neighbours reach FULL with a cost you can calculate from reference bandwidth over interface bandwidth.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 03 · HANDS-ON

Configure Multiarea OSPFv2 with Summarization and Filtering

LAB 16

Configure simple OSPFv2 environments including multiple normal areas, summarization and filtering (ENCOR 3.2.b).

The campus OSPF domain has grown to a single area carrying 400 routes, and every link flap triggers a full SPF run everywhere. As the engineer you split it into multiple areas, place the ABR correctly, summarise at the area boundary to contain the flooding, and make LAN interfaces passive so no adjacency is ever attempted towards a host.

YOU'LL BUILD

A multiarea OSPFv2 domain with area 0 backbone plus areas 1 and 2, route summarization at the ABR, passive LAN interfaces, and a stub area that blocks external LSAs.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1/R2/R3 routers, PC1/PC2/PC3

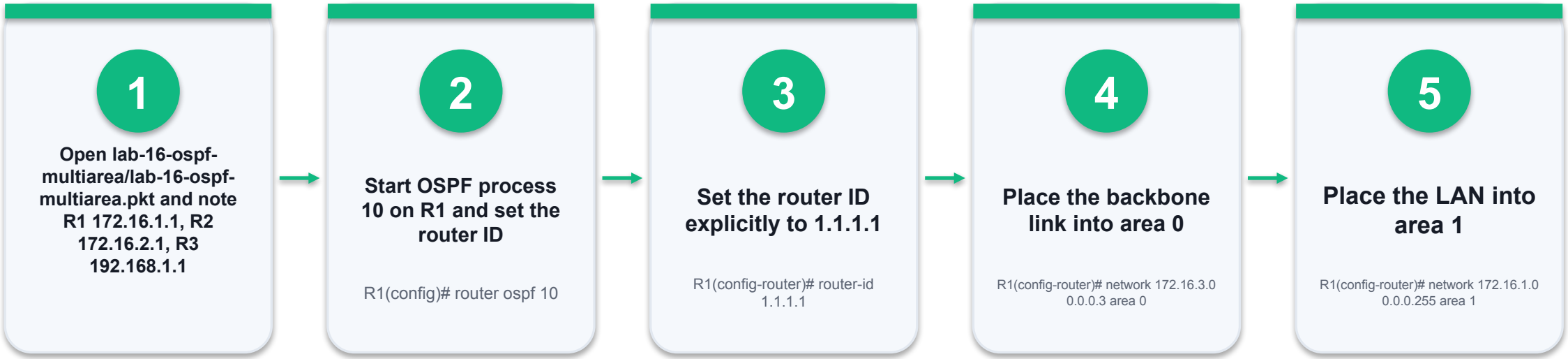
DONE WHEN

All OSPF neighbours are FULL, 'show ip ospf' identifies the ABR with interfaces in two areas, the area 1 prefixes appear as one summarised Type 3 LSA in the backbone, and the stub area's database contains no Type 5 external LSAs.

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LAB 16 · PROCESS MAP

How Lab 16 Runs



THE POINT

Configure simple OSPFv2 environments including multiple normal areas, summarization and filtering (ENCOR 3.2.b).

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LAB 16 · VERIFY

Configure Multiarea OSPFv2 with Summarization and Filtering

✅ Expected result

All OSPF neighbours are FULL, 'show ip ospf' identifies the ABR with interfaces in two areas, the area 1 prefixes appear as one summarised Type 3 LSA in the backbone, and the stub area's database contains no Type 5 external LSAs.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 03 · HANDS-ON

Configure Multiarea OSPFv3 for IPv6

LAB 17

Configure simple OSPFv3 environments including multiple normal areas (ENCOR 3.2.b).

The organisation is dual-stacking, and IPv6 needs its own routing. As the engineer you configure OSPFv3 across three routers, working with the differences that catch people out: OSPFv3 is enabled per interface rather than with network statements, it still needs a 32-bit router ID even though it routes IPv6, and adjacencies form over link-local addresses.

YOU'LL BUILD

A multiarea OSPFv3 domain with RA in area 1, RB in area 0 and RC in area 2, full adjacencies over link-local addresses, and end-to-end IPv6 reachability.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, RA/RB/RC routers

DONE WHEN

'show ipv6 ospf neighbor' shows all neighbours FULL with link-local addresses as the neighbour address, 'show ipv6 route ospf' contains OI inter-area routes, and RA pings RC's LAN across the backbone.

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LAB 17 · PROCESS MAP

How Lab 17 Runs



THE POINT

Configure simple OSPFv3 environments including multiple normal areas (ENCOR 3.2.b).

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LAB 17 · VERIFY

Configure Multiarea OSPFv3 for IPv6

✅ Expected result

'show ipv6 ospf neighbor' shows all neighbours FULL with link-local addresses as the neighbour address, 'show ipv6 route ospf' contains OI inter-area routes, and RA pings RC's LAN across the backbone.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 03 · HANDS-ON

Configure and Verify eBGP

LAB 18

Configure and verify eBGP between directly connected neighbours, including best path selection and neighbour relationships (ENCOR 3.2.c).

ACME is multihoming to two ISP entry points for internet resilience. As the WAN engineer you configure eBGP to both providers, advertise your own prefix, then influence which path inbound and outbound traffic takes — and you deliberately break the session by peering to a loopback without ebgp-multihop, so you recognise the most common eBGP failure in the field.

YOU'LL BUILD

Established eBGP sessions from ACME1 to both ISP entry points, the local prefix advertised and visible in both providers' tables, and a demonstrated path preference change.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, ACME1, OtherCo1, ISP-Entry-1, ISP-Entry-2

DONE WHEN

'show ip bgp summary' shows both neighbours Established with a non-zero prefix count, 'show ip bgp' marks the best path with >, and after the local preference change the best-path marker moves to the preferred provider's route.

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LAB 18 · PROCESS MAP

How Lab 18 Runs

1

Open lab-18-ebgp/lab-18-ebgp.pkt and confirm ACME1 S0/0/0 is 1.1.1.2 peering to 1.1.1.1

ACME1# ping 1.1.1.1

→

2

Start BGP with ACME's autonomous system number

ACME1(config)# router bgp 65001

→

3

Define the eBGP neighbour and its remote AS

ACME1(config-router)# neighbor 1.1.1.1 remote-as 65000

→

4

Advertise the local LAN prefix into BGP

ACME1(config-router)# network 192.168.0.0 mask 255.255.255.0

→

5

Disable auto-summary so classful summarisation does not surprise you

ACME1(config-router)# no auto-summary

THE POINT

Configure and verify eBGP between directly connected neighbours, including best path selection and neighbour relationships (ENCOR 3.2.c).

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LAB 18 · VERIFY

Configure and Verify eBGP

✅ Expected result

'show ip bgp summary' shows both neighbours Established with a non-zero prefix count, 'show ip bgp' marks the best path with >, and after the local preference change the best-path marker moves to the preferred provider's route.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 03 · HANDS-ON

Configure a Basic WLAN and WPA2-Enterprise on the WLC

LAB 19

Describe AP discovery and join, wireless segmentation with groups, profiles and tags, and configure wireless security features (ENCOR 3.3.c, 3.3.f, 5.4).

The office is deploying its first controller-based wireless network. As the wireless engineer you build the WLC configuration end to end through the GUI: interfaces and VLANs, a PSK guest WLAN, then a WPA2-Enterprise corporate WLAN backed by RADIUS. You watch the AP complete its CAPWAP discovery and join, and confirm clients land in the right VLAN.

YOU'LL BUILD

A WLC with a management interface on 192.168.200.254, a joined LAP, a PSK guest WLAN and a WPA2-Enterprise corporate WLAN authenticating against the RADIUS server at 172.31.1.254.

TOOLCHAIN

Cisco Packet Tracer, WLC-1 GUI, LAP-1, R1, SW1, RADIUS server, wireless hosts

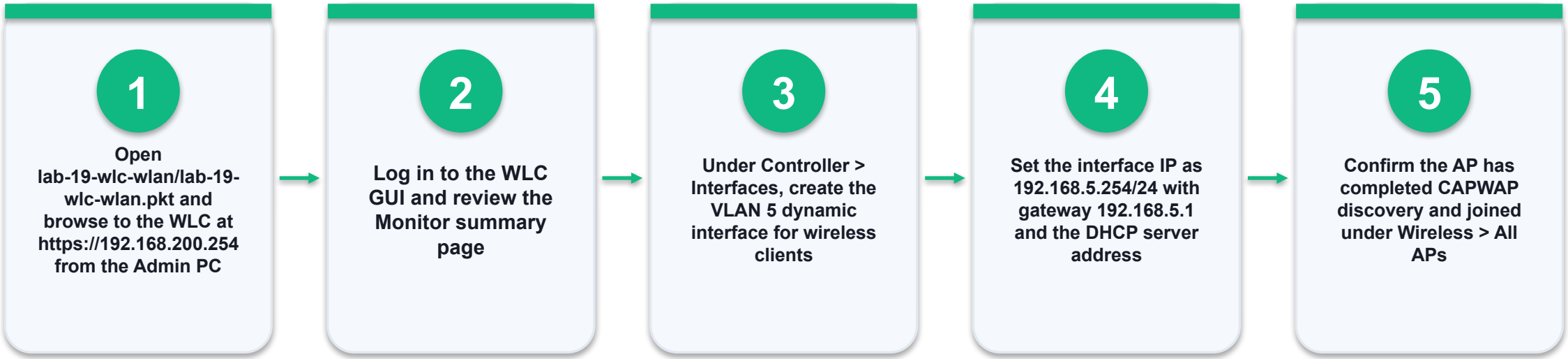
DONE WHEN

The LAP shows as joined and associated in Wireless > All APs, both WLANs appear enabled in the WLAN list, the guest client receives a 192.168.5.x address, and the corporate client authenticates against RADIUS and appears under Monitor > Clients.

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LAB 19 · PROCESS MAP

How Lab 19 Runs



THE POINT

Describe AP discovery and join, wireless segmentation with groups, profiles and tags, and configure wireless security features (ENCOR 3.3.c, 3.3.f, 5.4).

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LAB 19 · VERIFY

Configure a Basic WLAN and WPA2-Enterprise on the WLC

✅ Expected result

The LAP shows as joined and associated in Wireless > All APs, both WLANs appear enabled in the WLAN list, the guest client receives a 192.168.5.x address, and the corporate client authenticates against RADIUS and appears under Monitor > Clients.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 03 · HANDS-ON

Troubleshoot WLAN and Wireless Client Connectivity

LAB 20

Troubleshoot WLAN configuration and wireless client connectivity issues using GUI only (ENCOR 3.3.e).

Three separate wireless complaints arrive the same morning: one user cannot see the SSID, one associates but gets no IP address, and one authenticates but cannot reach the internal server. As the wireless support engineer you must resolve all three using the WLC GUI only — no CLI — which is exactly how the exam frames this objective.

YOU'LL BUILD

Three resolved wireless faults with a documented root cause for each: a disabled/misbroadcast WLAN, a wrong interface-to-VLAN mapping, and a RADIUS shared-secret mismatch.

TOOLCHAIN

Cisco Packet Tracer, WLC-1 GUI, LAP-1, RADIUS server, Laptop1 and Laptop2

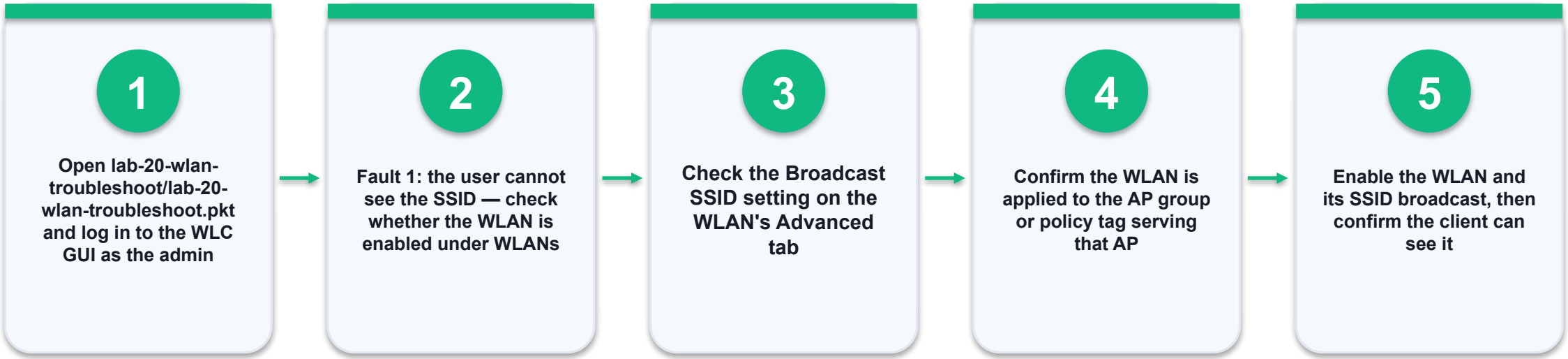
DONE WHEN

All three clients reach a working state: the SSID is visible and broadcast, the DHCP client holds an address in the correct VLAN, the 802.1X client authenticates, and troubleshooting-log.csv names a distinct root cause for each fault.

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LAB 20 · PROCESS MAP

How Lab 20 Runs



THE POINT

Troubleshoot WLAN configuration and wireless client connectivity issues using GUI only (ENCOR 3.3.e).

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LAB 20 · VERIFY

Troubleshoot WLAN and Wireless Client Connectivity

✅ Expected result

All three clients reach a working state: the SSID is visible and broadcast, the DHCP client holds an address in the correct VLAN, the 802.1X client authenticates, and troubleshooting-log.csv names a distinct root cause for each fault.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 03 · HANDS-ON

Configure NTP, NAT/PAT and HSRP

LAB 21

Interpret NTP configurations, configure NAT/PAT, and configure first hop redundancy protocols such as HSRP and VRRP (ENCOR 3.4.a, 3.4.b, 3.4.c).

The branch office needs three IP services before it goes live: synchronised time so logs correlate, PAT so its private hosts reach the internet through one public address, and a redundant default gateway so a router failure does not black-hole the LAN. As the branch engineer you configure all three and prove the HSRP failover actually works.

YOU'LL BUILD

A branch router synchronised to an NTP master, PAT translating 192.168.10.0/24 out one public address, and an HSRP group with virtual IP 192.168.1.254 that fails over on preemption.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1/R2/R3 routers, S1/S3 switches, PC-A and PC-B

DONE WHEN

'show ntp status' reports the clock is synchronised, 'show ip nat translations' lists inside-local to inside-global entries with differing ports, and 'show standby brief' shows R1 Active / R3 Standby — with R3 becoming Active and PC-A's ping continuing when R1's interface is shut.

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LAB 21 · PROCESS MAP

How Lab 21 Runs

1

Open lab-21-ip-services/lab-21-ip-services.pkt and review the three service requirements

2

Configure R1 as the NTP master at stratum 3

R1(config)# ntp master 3

3

Point R2 at R1 as its NTP server

R2(config)# ntp server 209.165.200.226

4

Verify the clock synchronised and note the stratum

R2# show ntp status

5

Check the association and confirm the * marks the synchronised peer

R2# show ntp associations

THE POINT

Interpret NTP configurations, configure NAT/PAT, and configure first hop redundancy protocols such as HSRP and VRRP (ENCOR 3.4.a, 3.4.b, 3.4.c).

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LAB 21 · VERIFY

Configure NTP, NAT/PAT and HSRP

✅ Expected result

'show ntp status' reports the clock is synchronised, 'show ip nat translations' lists inside-local to inside-global entries with differing ports, and 'show standby brief' shows R1 Active / R3 Standby — with R3 becoming Active and PC-A's ping continuing when R1's interface is shut.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC RECAP

Recap — Network Infrastructure

- You can now: Troubleshoot static and dynamic 802.1q trunking protocols (ENCOR 3.1.a).
- You can now: Troubleshoot static and dynamic EtherChannels — PAgP and LACP (ENCOR 3.1.b).
- You can now: Configure and verify common Spanning Tree Protocols — RSTP — and Spanning Tree enhancements (ENCOR 3.1.c).
- You can now: Configure and verify common Spanning Tree Protocols — MST (ENCOR 3.1.c).
- You can now: Configure and verify Spanning Tree enhancements such as root guard and BPDU guard (ENCOR 3.1.c).
- You can now: Compare routing concepts of EIGRP and OSPF — advanced distance vector versus link state, load balancing, path selection, metrics and area types (ENCOR 3.2.a).

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04

TOPIC 04

Network Assurance

Diagnostics · NetFlow · SPAN/RSPAN/ERSPAN · IP SLA · DNA Center · NETCONF and RESTCONF

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TOPIC 04 · SCOPE

Topic 04 — What This Domain Covers

1

Exam domain

ENCOR domain 4.0 Network Assurance

2

Exam weighting

10% of the ENCOR 350-401 exam

3

TSC mapping

Skills Framework K5, A4 · Network Configuration (ICT-DIT-3009-1.1)

4

Hands-on labs

Labs 22–25 — every objective has a lab

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TOPIC WEIGHTING 10%

Key Concepts — Network Assurance

1. Diagnose from the bottom up: ping proves reachability, extended ping proves it from the right source, traceroute locates where the path breaks, and debug shows why — but debug is CPU-expensive and must be scoped.

2. Conditional debugging ('debug interface', 'debug platform condition') restricts output to one interface, one MAC or one IP so you can safely debug on a production device without drowning the console.

3. SNMP polls devices for counters and state: v2c is community-string based and sends everything in clear text, while v3 adds authentication and encryption and is the only version acceptable in production.

4. Syslog messages carry a facility, a severity 0–7 (0 emergency to 7 debugging) and a mnemonic; sending them to a central collector with accurate NTP timestamps is what makes post-incident correlation possible.

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TOPIC WEIGHTING 10%

Key Concepts — Network Assurance (2)

1. Flexible NetFlow is built from three parts: a flow record (the key and non-key fields you want), a flow exporter (where to send it), and a flow monitor (which binds them and is applied to an interface).

2. NetFlow answers 'who is talking to whom, how much and for how long' — it is a summarised conversation record, not a packet capture, so it scales to a whole network where SPAN never could.

3. SPAN copies traffic from a source port or VLAN to a destination port on the same switch, RSPAN carries the copy across a dedicated VLAN to another switch, and ERSPAN encapsulates it in GRE so it can cross a routed network.

4. IP SLA generates synthetic probes — ICMP echo, UDP jitter, HTTP — and measures latency, jitter and loss continuously, so you learn the path degraded before the user calls.

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TOPIC WEIGHTING 10%

Key Concepts — Network Assurance (3)

1. Tracking an IP SLA operation with an object lets you drive real behaviour from measurement: a static route or an HSRP priority can be withdrawn automatically when the probe fails.

2. Cisco DNA Center organises work into four workflows — Design (site hierarchy and settings), Policy (SGTs and access), Provision (push to devices) and Assurance (health scores and issues) — and it is API-first, so anything the GUI does can be automated.

3. NETCONF is an XML-based configuration protocol running over SSH on port 830 with explicit candidate/running datastores and transactional commits, so a partial failure rolls back cleanly.

4. RESTCONF exposes the same YANG models as HTTPS REST on port 443 using GET, POST, PUT, PATCH and DELETE with JSON or XML — simpler for scripts, but without NETCONF's transactional guarantees.

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WHAT YOU'LL DO

Hands-On Labs — Network Assurance

Labs 22–23

- Diagnose Network Problems with Ping, Traceroute and Conditional Debug
- Configure Syslog, SNMP and Flexible NetFlow

Labs 24–25

- Configure SPAN, RSPAN and IP SLA
- Configure NETCONF and RESTCONF and Explore DNA Center Workflows

—

- —

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TOPIC 04 · HANDS-ON

Diagnose Network Problems with Ping, Traceroute and Conditional Debug

LAB 22

Diagnose network problems using tools such as debugs, conditional debugs, traceroute, ping, SNMP and syslog (ENCOR 4.1).

A user reports the application server is 'sometimes slow and sometimes down'. As the engineer you work the problem methodically from the bottom of the stack up: extended ping from the right source with the right size, traceroute to locate where the path breaks, then a conditional debug scoped to one host so you can safely watch the control plane on a production router.

YOU'LL BUILD

A completed diagnostic record locating the failure to a specific hop, with an extended ping result set, a traceroute trace and safely-scoped conditional debug output.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1/R2/R3 routers, PC-A

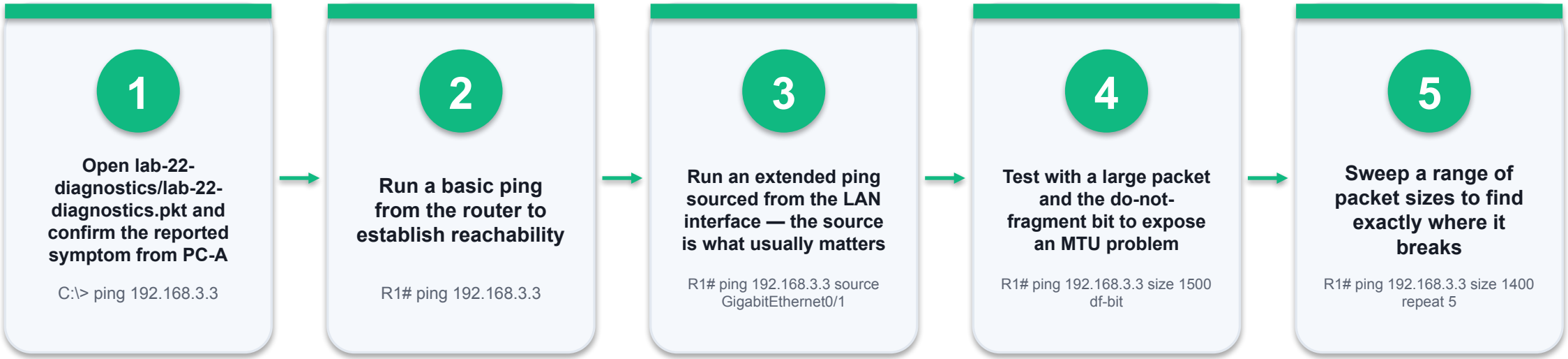
DONE WHEN

Your diagnostic record names the exact hop where the path fails, the sweep identifies the largest packet size that succeeds, and 'undebug all' confirms no debug is left running on the device.

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LAB 22 · PROCESS MAP

How Lab 22 Runs



THE POINT

Diagnose network problems using tools such as debugs, conditional debugs, traceroute, ping, SNMP and syslog (ENCOR 4.1).

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LAB 22 · VERIFY

Diagnose Network Problems with Ping, Traceroute and Conditional Debug

✅ Expected result

Your diagnostic record names the exact hop where the path fails, the sweep identifies the largest packet size that succeeds, and 'undebug all' confirms no debug is left running on the device.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 04 · HANDS-ON

Configure Syslog, SNMP and Flexible NetFlow

LAB 23

Configure and verify Flexible NetFlow, and use SNMP and syslog for diagnosis (ENCOR 4.1, 4.2).

The network has no visibility: when something breaks, nobody can say what changed or who was using the bandwidth. As the engineer you fix that in three layers — syslog to a central collector with accurate timestamps, SNMPv3 for polled state, and Flexible NetFlow to answer who talked to whom, how much and for how long.

YOU'LL BUILD

A router logging to a syslog collector at severity informational, an SNMPv3 user with authentication and encryption, and a Flexible NetFlow monitor applied to an interface producing a populated flow cache.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1/R2 routers, syslog server, SNMP server

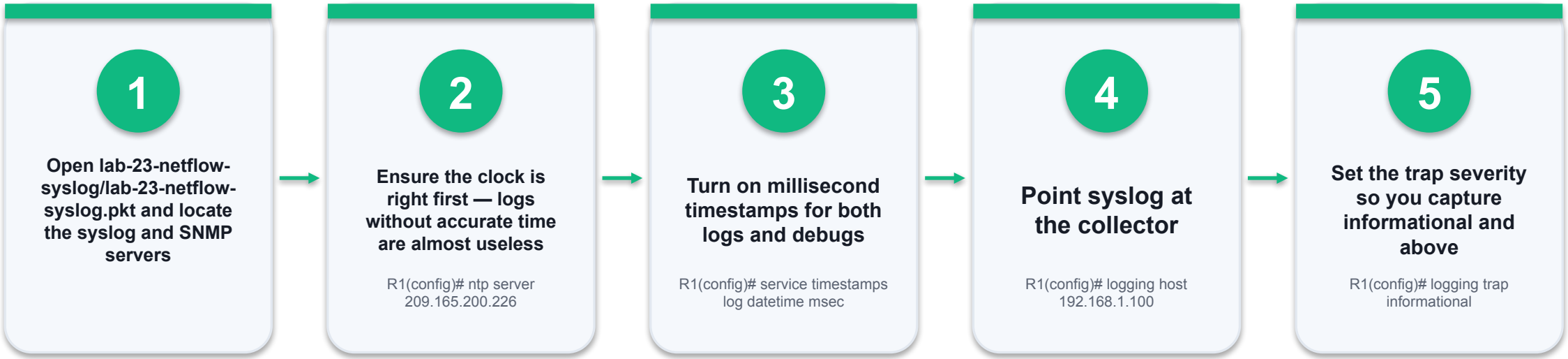
DONE WHEN

'show logging' confirms messages are being sent to 192.168.1.100 at informational severity, 'show snmp user' lists the v3 user with SHA and AES, and 'show flow monitor NETFLOW-MONITOR cache' displays flow entries with source, destination and byte counts.

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LAB 23 · PROCESS MAP

How Lab 23 Runs



THE POINT

Configure and verify Flexible NetFlow, and use SNMP and syslog for diagnosis (ENCOR 4.1, 4.2).

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LAB 23 · VERIFY

Configure Syslog, SNMP and Flexible NetFlow

✅ Expected result

'show logging' confirms messages are being sent to 192.168.1.100 at informational severity, 'show snmp user' lists the v3 user with SHA and AES, and 'show flow monitor NETFLOW-MONITOR cache' displays flow entries with source, destination and byte counts.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 04 · HANDS-ON

Configure SPAN, RSPAN and IP SLA

LAB 24

Configure SPAN/RSPAN/ERSPAN and configure and verify IPSLA (ENCOR 4.3, 4.4).

Security has asked for a packet capture of one server's traffic, and the WAN team wants to know when a path degrades before users complain. As the engineer you deliver both: SPAN to mirror the server port to an analyser, RSPAN to carry a copy to an analyser on another switch, and an IP SLA probe with a tracked object that withdraws a static route when the path fails.

YOU'LL BUILD

A working local SPAN session mirroring the server port, an RSPAN session across a dedicated VLAN, and an IP SLA ICMP echo operation with a tracked object driving a static route.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, S1/S2 switches, R1/R2 routers, analyser PC

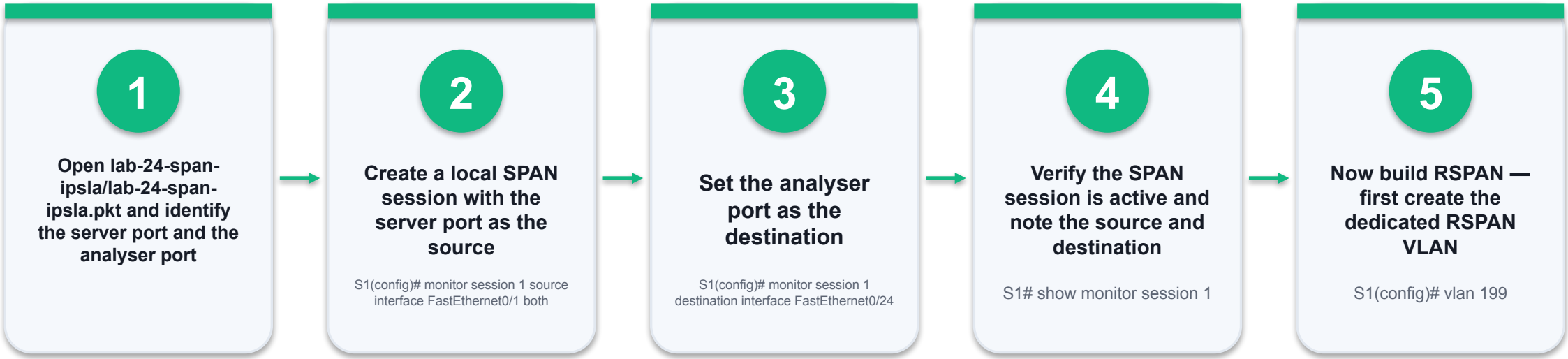
DONE WHEN

'show monitor session 1' shows an active local SPAN, the RSPAN session appears on both switches using VLAN 199, 'show ip sla statistics' reports successful probes with an RTT, and 'show track 1' reports the object Up with the default route present.

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LAB 24 · PROCESS MAP

How Lab 24 Runs



THE POINT

Configure SPAN/RSPAN/ERSPAN and configure and verify IPSLA (ENCOR 4.3, 4.4).

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LAB 24 · VERIFY

Configure SPAN, RSPAN and IP SLA

✅ Expected result

'show monitor session 1' shows an active local SPAN, the RSPAN session appears on both switches using VLAN 199, 'show ip sla statistics' reports successful probes with an RTT, and 'show track 1' reports the object Up with the default route present.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 04 · HANDS-ON

Configure NETCONF and RESTCONF and Explore DNA Center Workflows

LAB 25

Configure and verify NETCONF and RESTCONF, and describe Cisco DNA Center workflows (ENCOR 4.5, 4.6).

Your team is moving from CLI-by-hand to programmatic configuration. As the automation engineer you enable NETCONF and RESTCONF on the router, retrieve configuration through each, and compare them honestly: NETCONF's transactional datastores against RESTCONF's simpler REST verbs. You then map the four DNA Center workflows onto the tasks your team does manually today.

YOU'LL BUILD

A router with NETCONF-YANG on port 830 and RESTCONF on 443 both enabled and verified, a retrieved interface configuration through each protocol, and a completed DNA Center workflow mapping.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS XE CLI, Python 3, curl or Postman

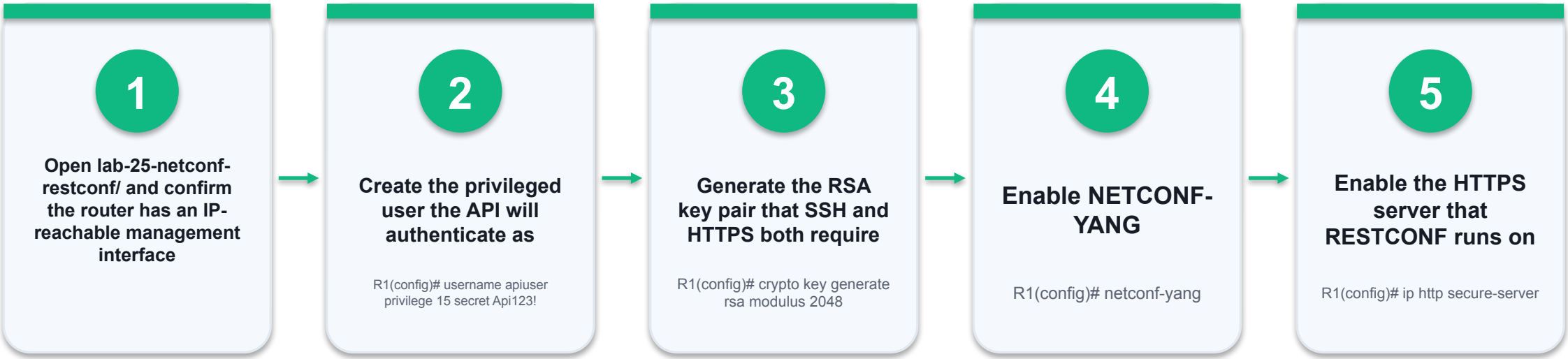
DONE WHEN

'show netconf-yang sessions' confirms the service is running, the curl call returns HTTP 200 with a JSON interface list, the Python NETCONF script returns the equivalent XML, and dnac-workflows.csv maps all four DNA Center workflows to real tasks.

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LAB 25 · PROCESS MAP

How Lab 25 Runs



THE POINT

Configure and verify NETCONF and RESTCONF, and describe Cisco DNA Center workflows (ENCOR 4.5, 4.6).

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LAB 25 · VERIFY

Configure NETCONF and RESTCONF and Explore DNA Center Workflows

✅ Expected result

'show netconf-yang sessions' confirms the service is running, the curl call returns HTTP 200 with a JSON interface list, the Python NETCONF script returns the equivalent XML, and dnac-workflows.csv maps all four DNA Center workflows to real tasks.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC RECAP

Recap — Network Assurance

- You can now: Diagnose network problems using tools such as debugs, conditional debugs, traceroute, ping, SNMP and syslog (ENCOR 4.1).
- You can now: Configure and verify Flexible NetFlow, and use SNMP and syslog for diagnosis (ENCOR 4.1, 4.2).
- You can now: Configure SPAN/RSPAN/ERSPAN and configure and verify IPSLA (ENCOR 4.3, 4.4).
- You can now: Configure and verify NETCONF and RESTCONF, and describe Cisco DNA Center workflows (ENCOR 4.5, 4.6).

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05

TOPIC 05

Network Security

Device access control · AAA · ACLs and CoPP · REST API security · wireless security · security design

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TOPIC 05 · SCOPE

Topic 05 — What This Domain Covers

1

Exam domain

ENCOR domain 5.0 Security

2

Exam weighting

20% of the ENCOR 350-401 exam

3

TSC mapping

Skills Framework K4, A6 · Network Configuration (ICT-DIT-3009-1.1)

4

Hands-on labs

Labs 26–30 — every objective has a lab

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TOPIC WEIGHTING 20%

Key Concepts — Network Security

1. Device access control starts with the lines: console, VTY and AUX each need a password and a login method, and 'transport input ssh' plus 'no ip http server' removes the plaintext paths.

2. Local user authentication with 'username … secret' stores a strong hash (type 8/9 scrypt where supported), whereas 'password' stores a trivially reversible type 7 — never use type 7 for anything that matters.

3. AAA separates Authentication (who are you), Authorization (what may you do) and Accounting (what did you do) — and 'aaa authentication login default group tacacs+ local' keeps a local fallback so a server outage does not lock you out.

4. TACACS+ encrypts the whole payload, uses TCP 49 and separates authentication from authorization, which is why it is preferred for device administration; RADIUS uses UDP, encrypts only the password, and is the standard for network access (802.1X).

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TOPIC WEIGHTING 20%

Key Concepts — Network Security (2)

1. A standard ACL matches source address only and belongs close to the destination; an extended ACL matches source, destination, protocol and port and belongs close to the source so unwanted traffic is dropped early.

2. Every ACL ends in an implicit 'deny any' — an ACL with only permit statements for your management subnet will silently drop everything else, which is the single most common cause of a self-inflicted lockout.

3. Control Plane Policing protects the CPU by applying a QoS policy to the control plane itself, rate-limiting management and routing traffic so a flood of ICMP or SNMP cannot starve the device of the CPU it needs to keep routing.

4. REST API security rests on transport (always HTTPS/TLS), authentication (tokens or OAuth 2.0 rather than embedded credentials), authorization scope, rate limiting, and input validation — and tokens must never be committed to a repository.

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TOPIC WEIGHTING 20%

Key Concepts — Network Security (3)

1. 802.1X is a three-party conversation: the supplicant (client), the authenticator (switch or WLC) and the authentication server (RADIUS/ISE); EAPOL carries EAP between supplicant and authenticator, RADIUS carries it onward.

2. WPA2/WPA3 personal uses a pre-shared key, enterprise uses 802.1X; either way the 4-way handshake derives fresh session keys (PTK/GTK) from the PMK, and it is what proves both sides hold the key without ever sending it.

3. WebAuth intercepts the first HTTP request and presents a captive portal — right for guest access, wrong for anything that needs strong identity, and useless for headless devices.

4. MAB (MAC Authentication Bypass) authenticates a device by its MAC address for printers and IP cameras that cannot run a supplicant — it is weak, because a MAC is trivially spoofed, so it is paired with profiling.

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TOPIC WEIGHTING 20%

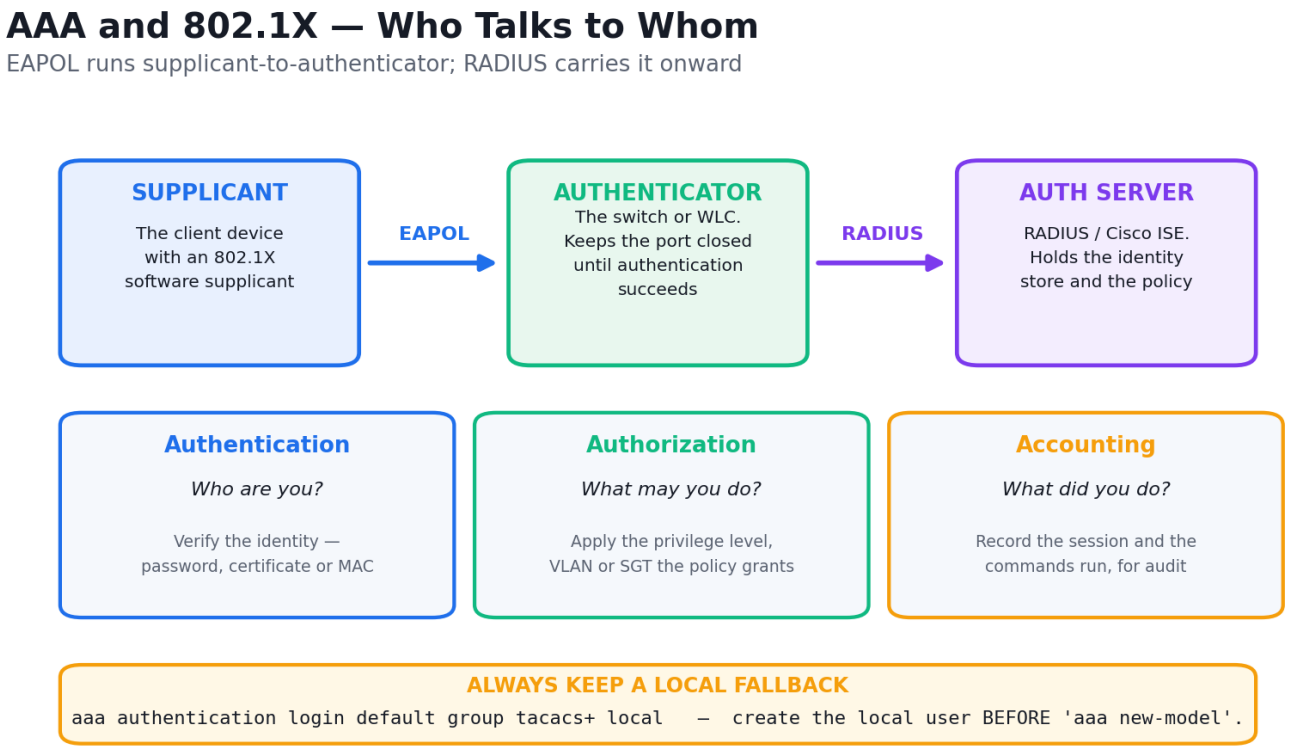
Key Concepts — Network Security (4)

1. Network security design layers threat defence (IPS/NGIPS), endpoint security (AMP), a next-generation firewall (application-aware, identity-aware), and segmentation with TrustSec SGTs and MACsec link encryption.

2. TrustSec tags traffic with a Security Group Tag at ingress and enforces it as an SGACL at egress, so policy follows identity rather than IP address — which is what makes policy survive DHCP and mobility.

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AAA and 802.1X — Who Talks to Whom



EAPOL runs between supplicant and authenticator; RADIUS carries it onward. Always create the local fallback user BEFORE 'aaa new-model'.

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WHAT YOU'LL DO

Hands-On Labs — Network Security

Labs 26–27

- Configure Secure Device Access with SSH and Strong Passwords
- Implement Local and Server-Based AAA

Labs 28–29

- Configure Extended ACLs to Mitigate Attacks
- Configure Control Plane Policing

Lab 30

- Analyse Wireless Security, REST API Security and Security Design

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TOPIC 05 · HANDS-ON

Configure Secure Device Access with SSH and Strong Passwords

LAB 26

Configure and verify device access control — lines and local user authentication (ENCOR 5.1.a).

An audit found the branch routers reachable by Telnet with a type 7 password that anyone can decode in seconds. As the engineer you harden device access: strong hashed local credentials, SSH version 2 only, Telnet removed from the VTY lines, an exec timeout, and login blocking so brute force attempts are throttled.

YOU'LL BUILD

A router and switch reachable only by SSHv2 with type 9 hashed local credentials, Telnet disabled, an exec timeout set and login-attempt throttling armed.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, RTA router, SW1 switch, PCA

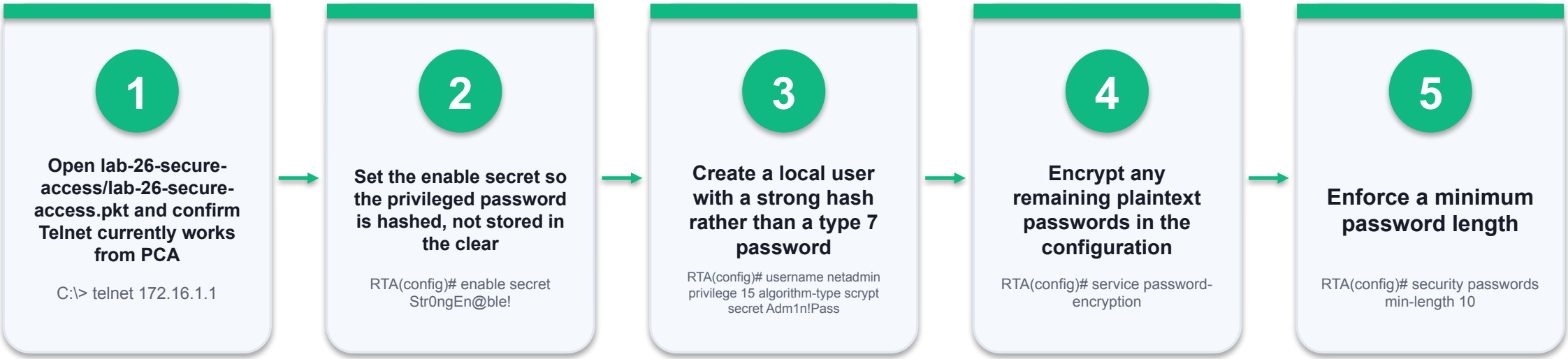
DONE WHEN

'show ip ssh' reports SSH Enabled version 2.0, Telnet from PCA is refused, SSH with the netadmin account succeeds, and 'show running-config | include username' shows a strong hash rather than a type 7 password.

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LAB 26 · PROCESS MAP

How Lab 26 Runs



THE POINT

Configure and verify device access control — lines and local user authentication (ENCOR 5.1.a).

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LAB 26 · VERIFY

Configure Secure Device Access with SSH and Strong Passwords

✅ Expected result

'show ip ssh' reports SSH Enabled version 2.0, Telnet from PCA is refused, SSH with the netadmin account succeeds, and 'show running-config | include username' shows a strong hash rather than a type 7 password.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 05 · HANDS-ON

Implement Local and Server-Based AAA

LAB 27

Configure and verify device access control — authentication and authorization using AAA (ENCOR 5.1.b).

Managing local accounts on 200 devices does not scale, and when an administrator leaves nobody can be sure every account is gone. As the engineer you centralise authentication on TACACS+ and RADIUS with AAA — and you deliberately keep a local fallback, because the classic AAA disaster is locking the whole team out when the server is unreachable.

YOU'LL BUILD

Routers authenticating administrators against TACACS+ (192.168.2.2) and RADIUS (192.168.3.2) with a working local fallback, plus authorization and accounting enabled.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1/R2/R3 routers, TACACS+ server, RADIUS server

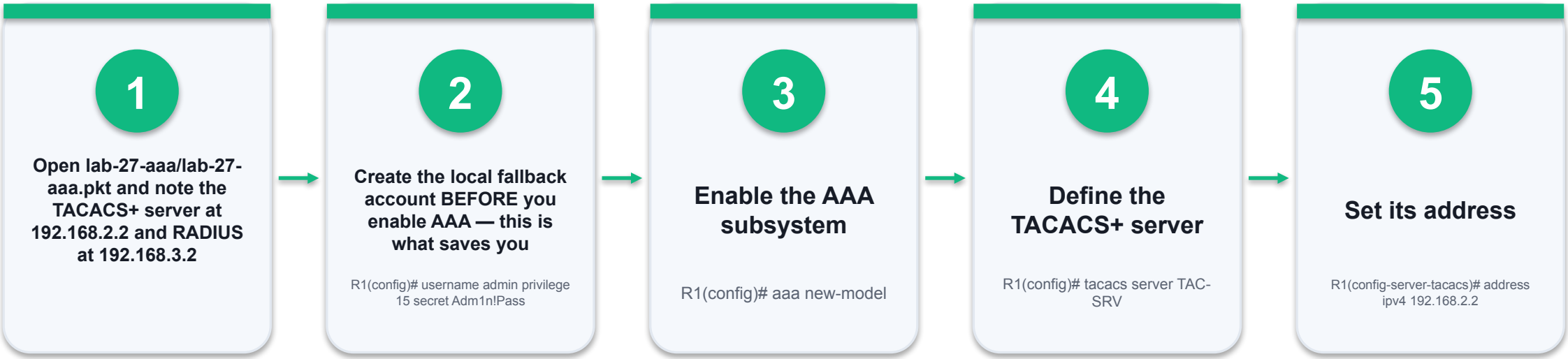
DONE WHEN

'test aaa group tacacs+' returns a successful authentication, 'show tacacs' shows the server reachable, and with the server link shut you can still log in with the local admin account — proving the fallback works.

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LAB 27 · PROCESS MAP

How Lab 27 Runs



THE POINT

Configure and verify device access control — authentication and authorization using AAA (ENCOR 5.1.b).

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LAB 27 · VERIFY

Implement Local and Server-Based AAA

✅ Expected result

'test aaa group tacacs+' returns a successful authentication, 'show tacacs' shows the server reachable, and with the server link shut you can still log in with the local admin account — proving the fallback works.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 05 · HANDS-ON

Configure Extended ACLs to Mitigate Attacks

LAB 28

Configure and verify infrastructure security features — ACLs (ENCOR 5.2.a).

The security team wants the branch to permit only what the business needs and to block the traffic patterns behind the last incident: spoofed private source addresses arriving from the internet and unsolicited inbound ICMP. As the engineer you write named extended ACLs, place them correctly, and confirm the implicit deny is not silently dropping something you needed.

YOU'LL BUILD

Named extended ACLs permitting only required services, blocking spoofed RFC 1918 sources inbound and unsolicited ICMP, applied in the correct direction with verified hit counters.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1/R2/R3 routers, PC-A, PC-C, servers

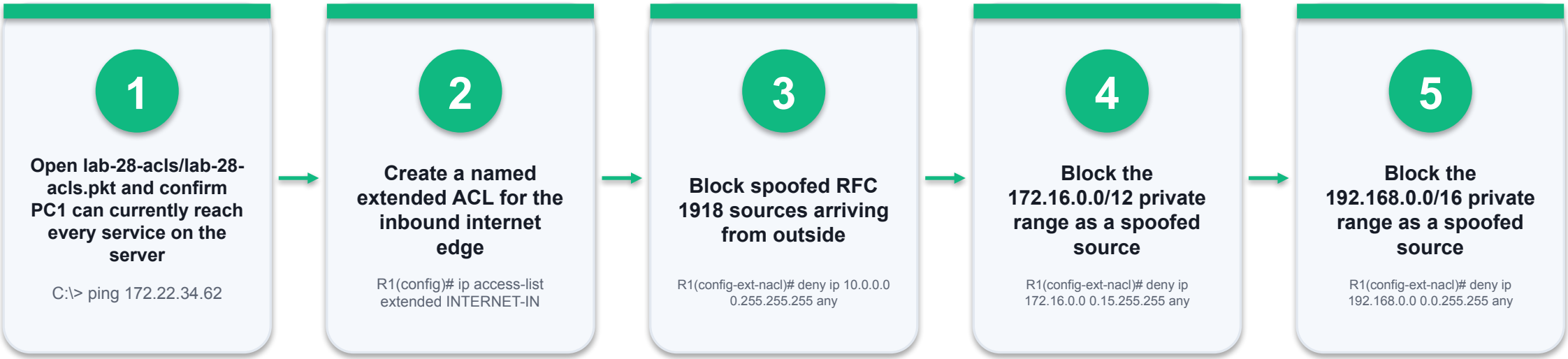
DONE WHEN

'show access-lists INTERNET-IN' shows the entries in the intended order with match counters incrementing on the permit lines, spoofed private-source traffic is denied, permitted HTTP/HTTPS still reaches the server, and the logged deny shows what the implicit deny would have dropped.

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LAB 28 · PROCESS MAP

How Lab 28 Runs



THE POINT

Configure and verify infrastructure security features — ACLs (ENCOR 5.2.a).

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LAB 28 · VERIFY

Configure Extended ACLs to Mitigate Attacks

✅ Expected result

'show access-lists INTERNET-IN' shows the entries in the intended order with match counters incrementing on the permit lines, spoofed private-source traffic is denied, permitted HTTP/HTTPS still reaches the server, and the logged deny shows what the implicit deny would have dropped.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 05 · HANDS-ON

Configure Control Plane Policing

LAB 29

Configure and verify infrastructure security features — CoPP (ENCOR 5.2.b).

During the last incident a flood of ICMP and SNMP drove the router CPU to 99% and OSPF adjacencies dropped — the device stopped routing because it was too busy answering management traffic. As the engineer you build a CoPP policy that rate-limits management and ICMP traffic to the control plane while guaranteeing routing protocol traffic always gets through.

YOU'LL BUILD

A CoPP policy applied to the control plane with separate classes for routing, management and ICMP, each policed at an appropriate rate, with verified conform and exceed counters.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1/R2 routers

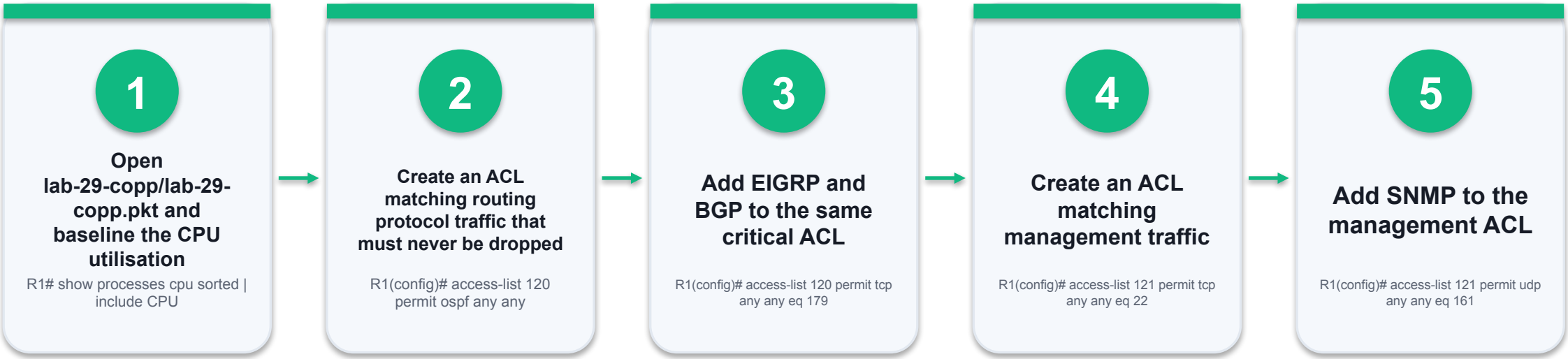
DONE WHEN

'show policy-map control-plane' shows the three classes with conformed bytes on all of them and exceeded/dropped bytes on the ICMP class during the flood, while 'show ip ospf neighbor' confirms the adjacency never dropped.

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LAB 29 · PROCESS MAP

How Lab 29 Runs



THE POINT

Configure and verify infrastructure security features — CoPP (ENCOR 5.2.b).

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LAB 29 · VERIFY

Configure Control Plane Policing

✅ Expected result

'show policy-map control-plane' shows the three classes with conformed bytes on all of them and exceeded/dropped bytes on the ICMP class during the flood, while 'show ip ospf neighbor' confirms the adjacency never dropped.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 05 · HANDS-ON

Analyse Wireless Security, REST API Security and Security Design

LAB 30

Describe REST API security, configure wireless security features, and describe the components of network security design (ENCOR 5.3, 5.4, 5.5).

You are reviewing three security designs before sign-off: the wireless authentication model for a new floor, the API integration a vendor has proposed, and the overall segmentation design. As the engineer you evaluate each against the exam's criteria — walking the 802.1X and 4-way handshake exchanges, finding the flaws in the vendor's API design, and placing each defence layer.

YOU'LL BUILD

A completed wireless authentication comparison, an annotated 4-way handshake walkthrough, a REST API security review naming every flaw in the vendor proposal, and a layered security design map.

TOOLCHAIN

Cisco Packet Tracer, WLC GUI, Postman or curl, security review worksheets

DONE WHEN

Your wireless comparison recommends 802.1X for corporate, PSK or WebAuth for guest and MAB with profiling for headless devices; your 4-way handshake walkthrough has all four messages in order; and api-security-review.csv names the HTTP transport, the embedded credentials and the absent rate limiting as flaws with concrete fixes.

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LAB 30 · PROCESS MAP

How Lab 30 Runs

1. Open lab-30-security-design/ and read the three review packs provided
2. Compare the wireless authentication methods in wireless-security.csv — PSK, 802.1X, WebAuth and MAB
3. Identify the three 802.1X roles: supplicant, authenticator and authentication server
4. Note that EAPOL carries EAP between supplicant and authenticator, and RADIUS carries it onward
5. Walk the 4-way handshake: the AP sends ANonce, the client replies with SNonce and a MIC

THE POINT

Describe REST API security, configure wireless security features, and describe the components of network security design (ENCOR 5.3, 5.4, 5.5).

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LAB 30 · VERIFY

Analyse Wireless Security, REST API Security and Security Design

✅ Expected result

Your wireless comparison recommends 802.1X for corporate, PSK or WebAuth for guest and MAB with profiling for headless devices; your 4-way handshake walkthrough has all four messages in order; and api-security-review.csv names the HTTP transport, the embedded credentials and the absent rate limiting as flaws with concrete fixes.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC RECAP

Recap — Network Security

- You can now: Configure and verify device access control — lines and local user authentication (ENCOR 5.1.a).
- You can now: Configure and verify device access control — authentication and authorization using AAA (ENCOR 5.1.b).
- You can now: Configure and verify infrastructure security features — ACLs (ENCOR 5.2.a).
- You can now: Configure and verify infrastructure security features — CoPP (ENCOR 5.2.b).
- You can now: Describe REST API security, configure wireless security features, and describe the components of network security design (ENCOR 5.3, 5.4, 5.5).

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06

TOPIC 06

Network Automation

Python · JSON · YANG · DNA Center and vManage APIs · REST responses · EEM · orchestration tools

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Topic 06 — What This Domain Covers

1

Exam domain

ENCOR domain 6.0 Automation

2

Exam weighting

15% of the ENCOR 350-401 exam

3

TSC mapping

Skills Framework K6, A5 · Network Configuration (ICT-DIT-3009-1.1)

4

Hands-on labs

Labs 31–35 — every objective has a lab

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TOPIC WEIGHTING 15%

Key Concepts — Network Automation

1. The ENCOR exam asks you to interpret Python, not to write an application: you must read a script and predict its output, so lists, dictionaries, loops, conditionals, functions and the requests library are the working set.

2. A Python dictionary maps keys to values and is exactly how a parsed JSON object arrives — nested API responses are therefore navigated with chained keys and indexes such as response['response'][0]['hostname'].

3. JSON has six value types — string, number, boolean, null, array and object — with keys always in double quotes, no trailing commas and no comments; a single trailing comma is the most common invalid-JSON exam trap.

4. YANG is a data modelling language that defines the structure, types and constraints of configuration and operational data; NETCONF and RESTCONF are the protocols that carry data conforming to those models.

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TOPIC WEIGHTING 15%

Key Concepts — Network Automation (2)

1. A YANG model is a tree of containers, lists, leaves and leaf-lists — a container groups, a list has a key and repeats, a leaf holds one typed value — and the same model drives both NETCONF XML and RESTCONF JSON.

2. The Cisco DNA Center Intent API is token-based: POST to /dna/system/api/v1/auth/token with Basic auth to receive an X-Auth-Token, then send that token in the header of every subsequent call.

3. The vManage API authenticates with a session cookie from /j_security_check plus an XSRF token for write operations, and exposes device inventory, statistics and policy over /dataservice endpoints.

4. REST response codes group by class: 2xx succeeded (200 OK, 201 Created, 204 No Content), 4xx you made an error (400 bad request, 401 unauthenticated, 403 unauthorised, 404 not found, 429 rate-limited), and 5xx the server failed.

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TOPIC WEIGHTING 15%

Key Concepts — Network Automation (3)

1. A 401 means the token is missing, wrong or expired, while a 403 means the token is valid but the account lacks permission — reading that difference correctly is the fastest way to fix a failing API script.

2. EEM (Embedded Event Manager) runs on the device itself: an event detector (syslog, SNMP, interface, timer, CLI) triggers an applet whose actions run CLI commands, write logs, send mail or set variables — automation with no external server.

3. An EEM applet needs 'event manager applet NAME', one event statement, and numbered action statements which run in lexical order — so 1.0, 2.0, 3.0 is the safe numbering convention, not 1, 2, 10.

4. Agent-based tools (Puppet, Chef) install software on the managed node and pull the desired state on a schedule; agentless tools (Ansible, SaltStack in agentless mode) push over SSH or an API and are far easier to adopt on network devices that cannot run an agent.

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TOPIC WEIGHTING 15%

Key Concepts — Network Automation (4)

1

Ansible is declarative and idempotent: a playbook describes the desired end state, and running it twice makes no second change — which is what makes it safe to re-run against production.

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WHAT YOU'LL DO

Hands-On Labs — Network Automation

Labs 31–32

- Interpret Python Components and Scripts
- Construct and Validate JSON-Encoded Files

Labs 33–34

- Explore YANG Models and Call the DNA Center API
- Construct an EEM Applet to Automate Response

Lab 35

- Compare Agent and Agentless Orchestration Tools

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TOPIC 06 · HANDS-ON

Interpret Python Components and Scripts

LAB 31

Interpret basic Python components and scripts (ENCOR 6.1).

The ENCOR exam does not ask you to write an application — it shows you a script and asks what it prints. As the network engineer moving into automation, you work through the Python that actually appears in network scripts: lists and dictionaries holding device inventories, loops that iterate them, conditionals that branch on state, and functions that return values.

YOU'LL BUILD

A set of correctly predicted script outputs, a working device-inventory script that loops a list of dictionaries, and a written explanation of the one deliberate bug you fix.

TOOLCHAIN

Python 3.10+, VS Code or any editor, devices.py sample scripts

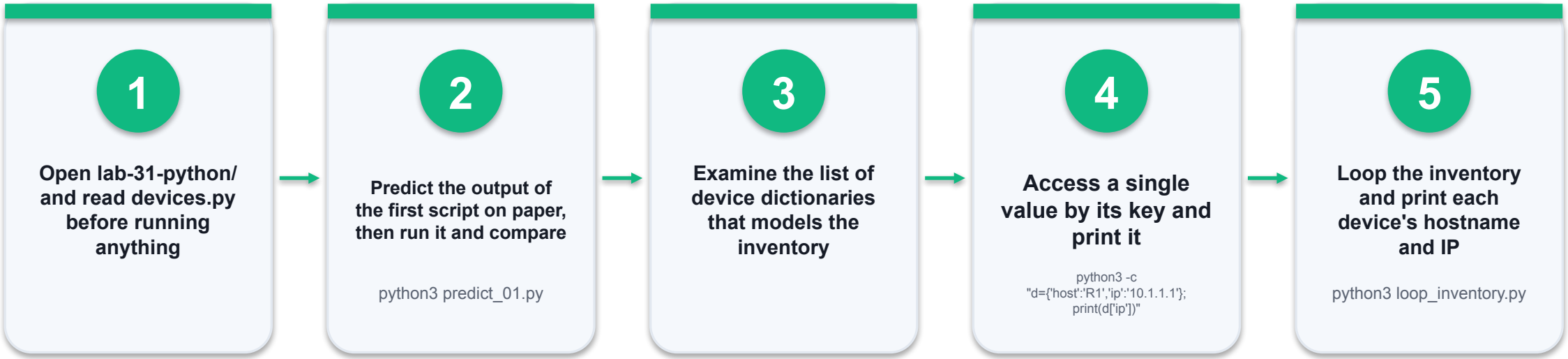
DONE WHEN

Your written predictions match the actual output of every script, inventory_report.py runs without error and prints one line per device, and buggy.py runs correctly after your fix with a comment explaining the indentation error.

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LAB 31 · PROCESS MAP

How Lab 31 Runs



THE POINT

Interpret basic Python components and scripts (ENCOR 6.1).

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LAB 31 · VERIFY

Interpret Python Components and Scripts

✅ Expected result

Your written predictions match the actual output of every script, inventory_report.py runs without error and prints one line per device, and buggy.py runs correctly after your fix with a comment explaining the indentation error.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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Construct and Validate JSON-Encoded Files

LAB 32

Construct valid JSON-encoded files (ENCOR 6.2).

Every REST API you will touch speaks JSON, and the exam presents you with JSON that is subtly invalid and asks you to spot why. As the automation engineer you construct a valid device inventory in JSON, deliberately break it in each of the classic ways, and learn to read a parser error back to the exact character that caused it.

YOU'LL BUILD

A valid nested JSON device inventory that passes validation, a set of four broken variants each demonstrating a different classic error, and a Python script that parses the valid file into a dictionary.

TOOLCHAIN

Python 3.10+, a text editor, json module, sample API payloads

DONE WHEN

devices.json validates cleanly with 'python3 -m json.tool', all four broken variants fail with a parser error you can explain, and your navigation command prints the correct nested interface name.

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LAB 32 · PROCESS MAP

How Lab 32 Runs

1. Open lab-32-json/ and review the six JSON value types — string, number, boolean, null, array, object
2. Write a JSON object describing one device with hostname, IP, uptime and an enabled flag
3. Extend it into an array of three device objects
4. Nest an interfaces array inside each device to make the structure realistic
5. Validate the file with Python's json module

```
python3 -m json.tool devices.json
```

THE POINT

Construct valid JSON-encoded files (ENCOR 6.2).

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LAB 32 · VERIFY

Construct and Validate JSON-Encoded Files

✅ Expected result

devices.json validates cleanly with 'python3 -m json.tool', all four broken variants fail with a parser error you can explain, and your navigation command prints the correct nested interface name.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 06 · HANDS-ON

Explore YANG Models and Call the DNA Center API

LAB 33

Describe the principles and benefits of YANG, describe APIs for Cisco DNA Center and vManage, and interpret REST API response codes (ENCOR 6.3, 6.4, 6.5).

Your team is scripting against Cisco DNA Center, and the first attempts are failing with codes nobody can interpret. As the automation engineer you walk the YANG model that defines the data, authenticate correctly to obtain an X-Auth-Token, then work through the response codes deliberately — including the 401 versus 403 distinction that decides whether you fix the token or the account's permissions.

YOU'LL BUILD

A YANG model walkthrough naming containers, lists and leaves, a successful token request, a device inventory retrieved with that token, and a completed response-code interpretation table.

TOOLCHAIN

Python 3.10+, requests library, curl or Postman, DNA Center sandbox or provided mock API

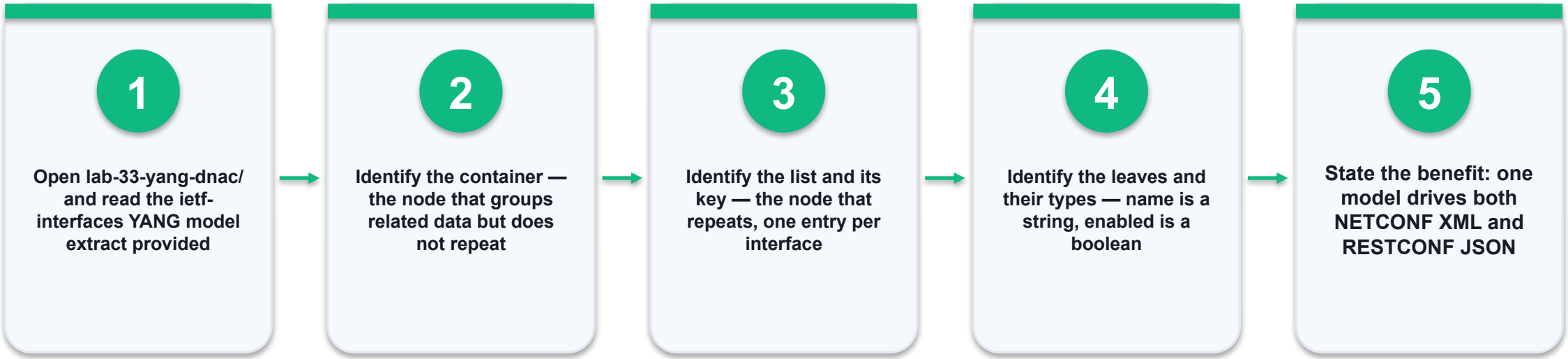
DONE WHEN

The token request returns 200 with a Token value, the authenticated inventory call returns 200 with a device list, the unauthenticated call returns 401, and response-codes.csv correctly distinguishes 401 (fix the token) from 403 (fix the permissions).

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LAB 33 · PROCESS MAP

How Lab 33 Runs



THE POINT

Describe the principles and benefits of YANG, describe APIs for Cisco DNA Center and vManage, and interpret REST API response codes (ENCOR 6.3, 6.4, 6.5).

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LAB 33 · VERIFY

Explore YANG Models and Call the DNA Center API

✅ Expected result

The token request returns 200 with a Token value, the authenticated inventory call returns 200 with a device list, the unauthenticated call returns 401, and response-codes.csv correctly distinguishes 401 (fix the token) from 403 (fix the permissions).

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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Construct an EEM Applet to Automate Response

LAB 34

Construct an EEM applet to automate configuration, troubleshooting or data collection (ENCOR 6.6).

A critical WAN interface flaps at night and nobody captures the state before it recovers, so the cause is never found. As the engineer you make the device investigate itself: an EEM applet that triggers on the syslog message, runs the diagnostic commands, writes the output to a file and logs a summary — automation with no external server involved.

YOU'LL BUILD

Two working EEM applets — one syslog-triggered applet that collects diagnostics on an interface down event, and one timed applet that logs a periodic health snapshot.

TOOLCHAIN

Cisco Packet Tracer, Cisco IOS CLI, R1 router

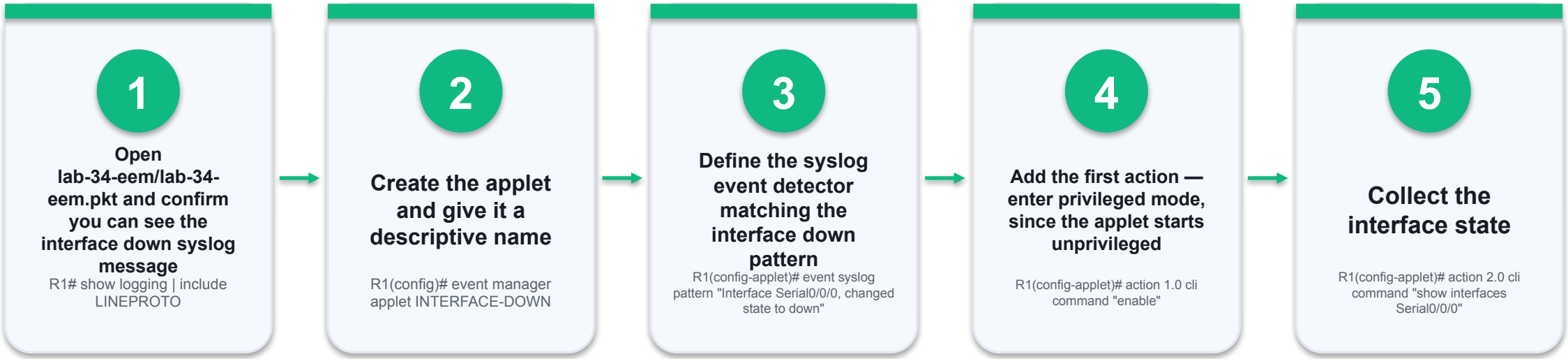
DONE WHEN

'show event manager policy registered' lists both applets, shutting Serial0/0/0 produces the 'EEM: Serial0/0/0 down' syslog entry, and 'show event manager statistics policy' shows a non-zero run count for the triggered applet.

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LAB 34 · PROCESS MAP

How Lab 34 Runs



THE POINT

Construct an EEM applet to automate configuration, troubleshooting or data collection (ENCOR 6.6).

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LAB 34 · VERIFY

Construct an EEM Applet to Automate Response

✅ Expected result

'show event manager policy registered' lists both applets, shutting Serial0/0/0 produces the 'EEM: Serial0/0/0 down' syslog entry, and 'show event manager statistics policy' shows a non-zero run count for the triggered applet.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC 06 · HANDS-ON

Compare Agent and Agentless Orchestration Tools

LAB 35

Compare agent versus agentless orchestration tools such as Chef, Puppet, Ansible and SaltStack (ENCOR 6.7).

Your organisation must pick one configuration management tool for a network of 300 Cisco devices that cannot run third-party agents. As the engineer making the recommendation you compare the four tools on architecture, transport, language and network-device suitability, then write and read a small Ansible playbook to prove the agentless model works on IOS.

YOU'LL BUILD

A completed four-tool comparison matrix, a written recommendation justified on the agent constraint, and an annotated Ansible playbook that configures an IOS interface idempotently.

TOOLCHAIN

Python 3.10+, Ansible, a text editor, tool comparison worksheet

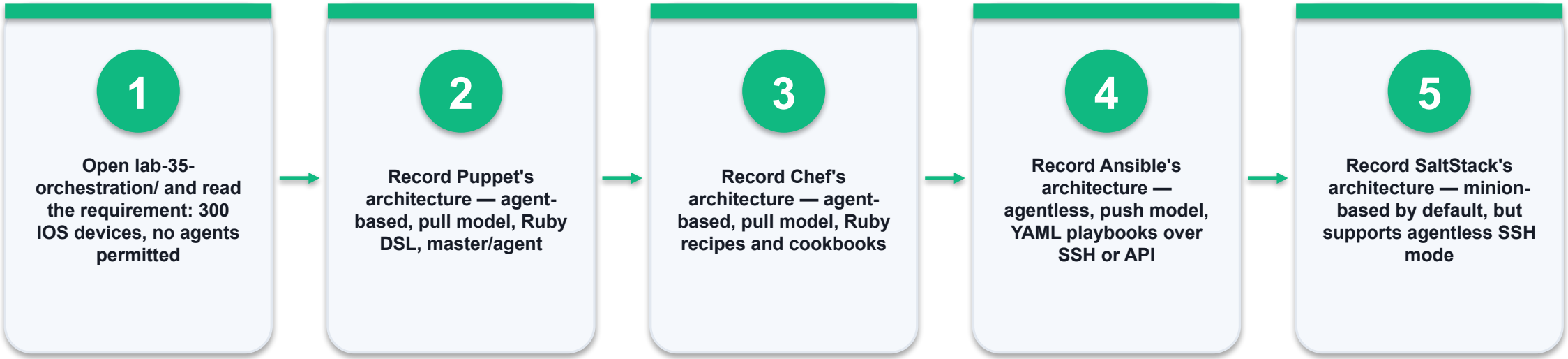
DONE WHEN

orchestration-comparison.csv correctly classifies Puppet and Chef as agent-based pull and Ansible as agentless push, the playbook reports changed=1 on the first run and changed=0 on the second, and your recommendation names Ansible with the no-agent constraint as the reason.

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LAB 35 · PROCESS MAP

How Lab 35 Runs



THE POINT

Compare agent versus agentless orchestration tools such as Chef, Puppet, Ansible and SaltStack (ENCOR 6.7).

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LAB 35 · VERIFY

Compare Agent and Agentless Orchestration Tools

✅ Expected result

orchestration-comparison.csv correctly classifies Puppet and Chef as agent-based pull and Ansible as agentless push, the playbook reports changed=1 on the first run and changed=0 on the second, and your recommendation names Ansible with the no-agent constraint as the reason.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm GOOGLE_GENAI_USE_VERTEXAI=0.

ModuleNotFoundError

Run uv sync again, and prefix commands with uv run so the venv is used.

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TOPIC RECAP

Recap — Network Automation

- You can now: Interpret basic Python components and scripts (ENCOR 6.1).
- You can now: Construct valid JSON-encoded files (ENCOR 6.2).
- You can now: Describe the principles and benefits of YANG, describe APIs for Cisco DNA Center and vManage, and interpret REST API response codes (ENCOR 6.3, 6.4, 6.5).
- You can now: Construct an EEM applet to automate configuration, troubleshooting or data collection (ENCOR 6.6).
- You can now: Compare agent versus agentless orchestration tools such as Chef, Puppet, Ansible and SaltStack (ENCOR 6.7).

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1 · Translated requirements

Turned organisational needs into campus, wireless and WAN designs — 2-tier versus 3-tier, deployment models and QoS policy.

2

LO2 · Selected components

Chose and configured the right components: VRFs, tunnels, switches, routers, controllers and their interdependencies.

3

LO3 · Implemented devices

Configured trunking, EtherChannel, RSTP, MST, EIGRP, OSPFv2/v3, eBGP, NAT, HSRP and controller-based wireless.

4

LO4 · Managed access

Secured device access with SSH and AAA, enforced ACLs and CoPP, and deployed 802.1X and WPA2-Enterprise wireless.

5

LO5 · Deployed tools

Deployed syslog, SNMPv3, Flexible NetFlow, SPAN/RSPAN, IP SLA, NETCONF and RESTCONF.

6

LO6 · Reported and assessed

Automated reporting and response with Python, JSON, REST APIs, EEM applets and Ansible.

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NEXT STEPS

Your Path to CCNP Enterprise

1

1 · Pass ENCOR 350-401

The core exam you have prepared for this week. Passing it alone earns Cisco Certified Specialist – Enterprise Core.

2

2 · Choose a concentration

CCNP Enterprise = ENCOR plus one concentration exam. ENARSI (advanced routing) is the most common next step.

3

3 · Practise the pacing

Use the practice exam at exams.tertiaryinfotech.com in Exam mode until 100 questions in 120 minutes feels comfortable.

4

4 · Rebuild every lab

Redo each of the 35 labs from a blank configuration until the commands come without the guide.

5

5 · Read the blueprint again

Walk learningnetwork.cisco.com/s/encor-exam-topics objective by objective and rate your confidence on each.

6

6 · Book the exam

Schedule at a Pearson VUE centre once you are consistently scoring above 85% in Exam mode.

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PRACTICE EXAM · exams.tertiaryinfotech.com

Practice Exam — Book Your Revision Now

1

Where

exams.tertiaryinfotech.com/practice-exams/cisco/cisco-ccnp-encor

2

What

200 original questions across two full 100-question practice exams.

3

How long

120 minutes each — the same as the real 350-401 exam.

4

Practice mode

Immediate answers and explanations. Use it while the material is still fresh.

5

Exam mode

Timed with answers hidden until submission. Use it in your final week.

6

Free sample

A free sample exam is available with no payment details required.

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CONTINUE WITH TERTIARY INFOTECH

Recommended Courses

1. WSQ Cisco Certified Network Associate (CCNA) Training
2. WSQ Cisco Certified Network Professional (CCNP) for ENARSI Training
3. WSQ Certified Ethical Hacker (CEH) Training
4. WSQ ISC2 Certified Information Systems Security Professional (CISSP) Training
5. WSQ CompTIA Network+ Training
6. WSQ Network Automation with Python and Ansible

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WE'RE HERE TO HELP

Support

- If you have any enquiries during or after the class, you can contact us below.
- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg
- LMS / TMS: https://lms-tms.tertiaryinfotech.com

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Final Assessment

1

Written Assessment (SAQ)

1 hour · open book · short-answer knowledge questions across all six domains.

2

Practical Performance (PP)

1 hour · open book · hands-on Cisco configuration and verification tasks.

3

Digital attendance

Remember to take the Assessment digital attendance (TRAQOM) before you start.

4

Submit on the LMS

Upload your completed answers at https://lms-tms.tertiaryinfotech.com/

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ON ASSESSMENT DAY

Assessment Flow

1

TRAQOM survey

Scan the QR code on the LMS

2

Digital attendance

Scan the SSG assessment QR

3

Sit WA then PP

Open book · 1 hour each

4

Submit on the LMS

Upload your completed answers

5

Sign the record

Sign the Assessment Summary Record

REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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GOOD LUCK IN THE EXAM

Thank You!

You can now design, implement, secure and automate the core technologies of a Cisco enterprise network — and you are ready to sit ENCOR 350-401.

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